A Conditional Reconstruction Program for Relativity and Standard Model Structure from Observer-Overlap Consistency

Observers Are All You Need (Compact)

B. Müller

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Abstract

We present a compact, falsifiable Phase-I statement of Observer-Patch Holography (OPH) as a conditional reconstruction program whose claimed recovered core is limited to relativity and Standard Model structure. From five axioms, the stated scaling-limit/categorical regularity package, and two external continuous inputs used only in the current quantitative implementation, the compact note derives a schedule-independent normal form for overlap consistency, a conditional Lorentz branch on the screen, a conditional Jacobson-type Einstein branch from fixed-cap generalized-entropy stationarity together with the null modular bridge and the scaling-limit null-stress premise, and conditional compact gauge reconstruction in the bosonic internal-gauge branch from a refinement-stable symmetric edge-sector colimit. After the extended Minimal Admissible Realization selector is imposed on the connected positive-dimensional Lie admissible class with one connected abelian charge factor, the realized low-energy gauge group on that realized branch is

$$\frac{\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_c = 3, \qquad N_g = 3,$$

together with the exact rational hypercharge lattice for the realized one-generation Standard Model chiral matter package with one Higgs doublet. The two external continuous inputs of the present implementation are the pixel area

$$P \equiv a_{\mathrm{cell}}/\ell_P^2 = 1.63094$$

and the total screen capacity

$$N_{\mathrm{scr}} \equiv \log\dim\mathcal{H}_{\mathrm{tot}} \sim 10^{122}.$$

In the current implementation, P feeds the supplement-backed gauge/electroweak calibration sector and N_{scr} feeds the capacity branch for Λ; the D10–D11 numerical branches are reported only as implementation-level checks, not as the recovered core itself. Charged-lepton/Koide, texture, benchmark neutrino-model, dark-sector, baryogenesis, black-hole-spectroscopy, and string/worldsheet discussions are explicitly deferred continuations requiring additional ansätze beyond the recovered core and are not claimed as outputs of overlap consistency plus the stated scaling-limit and categorical premises. The purpose of this compact note is therefore to state a clean Phase-I claim set: relativity and Standard Model structure are the recovered core, while all downstream branches are either input-dependent or supplement-backed secondary sectors, or post-Phase-I phenomenology.

1 Introduction

The standard model of particle physics and general relativity are successful effective descriptions. They leave open a familiar list of questions: why the Lorentz group appears, why the low-energy spacetime description follows a Jacobson-type Einstein branch, why the gauge structure is

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6},$$

why there are three colors and three generations, why hypercharges take the observed rational values, why the electroweak and flavor scales are what they are, and why dark-sector phenomenology and the cosmological constant enter in the particular form observed.

This paper presents a single framework that addresses this list as a conditional reconstruction program. The framework is Observer-Patch Holography (OPH). Its starting point is an observer-centric organization of physical data: each observer has access only to a patch algebra, neighboring observers must agree on overlaps, and global physical law is recovered from the resulting consistency conditions. The same formal structure that produces schedule-independent overlap consistency then feeds a Lorentzian modular-phase classification, a Jacobson-type Einstein branch conditional on fixed-cap stationarity and the null-stress premise, compact gauge reconstruction in a bosonic internal-gauge sector, charge quantization, and the low-energy architecture selected by the extended MAR admissibility package.

Two decisions shape the present version.

1. The compact note presents the main theorem statements in one place, but some quantitative conventions and implementation details live in repository supplements.[1]

2. The paper separates exact structural theorems, conditional scaling-limit branches, calibration-sector outputs, and phenomenological continuations. This keeps strong claims visible and weak claims identifiable.

The framework is intended as a candidate fundamental theory in the precise sense relevant for physics: general relativity and the Standard Model appear as effective sectors derived from a smaller axiom set plus an explicit refinement/scaling-limit package. The paper does not use simulation language, observer continuation stories, or other non-falsifiable additions.

Prioritized achievements

The results emphasized in this paper are ordered below by a combination of impact and defensibility.

1. A conditional Jacobson-type Einstein branch is recovered from observer-overlap modular geometry, an explicit fixed-cap generalized-entropy stationarity premise, and an explicit null-modular / null-stress scaling-limit package; in the synchronized c.10 bridge, quasi-local propagation and endpoint control are now supplied internally up to the weak-tail-generator stage, while half-sided inclusion and null-stress identification remain explicit downstream premises.

2. The exact Standard Model gauge structure

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}$$

[1] Questions about ultimate purpose are deferred until the corresponding physical question has been posed with enough precision to deserve an answer.

is recovered in the bosonic compact-gauge branch once the extended MAR admissibility package is imposed.

3. On the realized one-generation chiral matter plus one-Higgs package, anomaly constraints and Yukawa invariance fix the hypercharge lattice, and MAR then selects three colors and three generations on the realized admissible branch.

4. The current quantitative implementation uses two external continuous configuration inputs, P and N_{scr}.

5. The Higgs/top critical-surface branch adds no extra continuous fit once the gauge trajectory and scale-setting branch are fixed, while the charged-lepton/Koide continuation uses extra phenomenological structure beyond the core theorem package.

6. A separate input-dependent capacity corollary ties the cosmological-constant branch to global screen capacity rather than local vacuum energy.

7. Absence of gauge-mediated proton decay follows from product-group structure.

8. Dark-sector, baryogenesis, string, and spectroscopy branches are kept visible, but only as program-level or conjectural phenomenology.

9. Objective physical law is identified with the unique normal form of the overlap-consistency problem.

2 Results at a Glance

This section is the authoritative status ledger for the compact note and the sync rule for mirrored summary surfaces. The three-phase boundary used throughout is

$$\text{Phase I} = (D1\text{–}D5) \cup (D7\text{–}D9), \qquad \text{Phase II} = D6 \cup D10 \cup D11, \qquad \text{Phase III} = D12.$$

Phase I is the recovered-core claim set. Phase II contains the separate input-dependent capacity corollary and the current supplement-backed numerical implementation. Phase III collects deferred continuations. The canonical five-axiom basis, the two external continuous inputs, and the technical-premise labels used below are fixed in Section 3. Each row names a node in the reconstruction program, records its immediate OPH-side parents, separates imported standard mathematics from extra non-axiom physical premises or external inputs, and states the resulting claim tier.

Table 1: Authoritative dependency checklist for the compact OPH program.

Node	Output	Immediate OPH ingredients	Standard mathematics used	Extra non-axiom physical premises / external inputs	Claim tier
D1	Unique schedule-independent normal form (Theorem 4.3)	overlap-consistency problem on a finite patch net	Newman's lemma for terminating locally confluent rewrite systems	finite state spaces, termination, and local confluence	Phase I structural theorem

Table 1: Authoritative dependency checklist for the compact OPH program.

Node	Output	Immediate OPH ingredients	Standard mathematics used	Extra non-axiom physical premises / external inputs	Claim tier
D2	Collar Markov/entropy split and G-dictionary setup (Theorem 5.2, Propositions 5.6, 5.7, 5.10, Definition 5.11)	Axioms 3.1–3.4	HJPW Markov structure theorem; Petz and Fawzi–Renner recovery theory	exact block-factorized identities require exact Markovity or an explicitly stated idealized limit; the $A/(4G)$ step is the coarse-grained dictionary of Definition 5.11	Phase I structural lemma/proposition layer
D3	Lorentz branch $\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1)$ (Theorem 6.1, Corollary 6.3)	Axioms 3.1–3.4	Bisognano–Wichmann modular geometry; conformal-group classification	T8 together with the OPH geometric-branch hypothesis packaged in Theorem 6.1	Phase I conditional scaling-limit theorem
D4	Null modular bridge to T_{kk} (Propositions 6.4, 6.7; Corollaries 6.5, 6.9; Theorem 6.8, Lemma 6.10; Theorem 6.11)	D2+D3 and Axioms 3.1–3.4	Borchers–Wiesbrock half-sided modular inclusion; distributional differentiation on half-lines	the theorem-local inherited-strip decomposition and exact-or-controlled Markov hypotheses of Proposition 6.4 and Corollary 6.5, plus T6 and T8 together with the relativistic null-stress identification premise; the propagation controls denoted T2–T3 are supplied internally by Axiom 3.3 on the local finite-constraint branch	Phase I conditional bridge theorem
D5	Jacobson-type Einstein branch: rest-frame relation plus conditional tensor upgrade (Lemma 6.16, Theorem 6.17, Corollary 6.18)	D3+D4 and Axioms 3.1–3.4	Jacobson small-ball first law; null-to-tensor reconstruction modulo a metric term	T9 fixed-cap stationarity; locally Lorentzian $d = 4$ scaling regime; small-ball constancy assumptions; all local directions/reference states for the tensor upgrade	Phase I conditional scaling-limit theorem/corollary
D6	Capacity branch for Λ and the separate order-of-magnitude neutrino estimate (Proposition 6.20, Corollary 6.21)	D5 leaves a $+\Lambda g_{ab}$ ambiguity	de Sitter entropy relation and dimensional analysis	external input N_{scr} and the screen-capacity identification	Phase II input-dependent corollary / estimate
D7	Compact gauge reconstruction (Theorem 7.1)	Axioms 3.1–3.4	Doplicher–Roberts / Tannaka reconstruction	T7 and T10–T11 together with the directed-colimit/fiber clause of T12; the refinement-stable qualifier on that colimit is supplied internally by Axiom 3.3	Phase I conditional structural theorem
D8	Product gauge structure up to finite quotient (Lemmas 7.3–7.8, Theorem 7.9)	D7 + Axiom 3.5	compact Lie representation classification; Schur's lemma	connected positive-dimensional Lie admissible class; one connected abelian factor; faithful action on the minimal coupled carrier; vanishing obstruction $[z] = 0$	Phase I realized-branch theorem

Table 1: Authoritative dependency checklist for the compact OPH program.

Node	Output	Immediate OPH ingredients	Standard mathematics used	Extra non-axiom physical premises / external inputs	Claim tier
D9	Realized Standard Model quotient, hypercharges, $N_c = 3$, $N_g = 3$, and product-group corollaries (Theorem 7.10, Corollaries 7.11, 7.12, Proposition 7.13)	D8	anomaly-cancellation algebra; Witten global-anomaly argument; CKM CP counting	realized one-generation chiral matter plus one Higgs package; asymptotic-freedom and MAR admissibility premises	Phase I realized-branch theorem/corollary chain
D10	Gauge/electroweak numerical closure of the current supplement-backed implementation	D9 + external input P	RG evolution and matching conventions supplied in the supplement	pixel constraint; edge heat-kernel closure; supplement-backed beta-function and threshold conventions	Phase II calibration sector
D11	Higgs/top critical-surface branch of the current supplement-backed implementation	D10	RG transport and pole-mass conversion supplied in the supplement	critical-surface condition $\lambda = \beta_\lambda = 0$ at M_U; no new continuous fit once D10 is fixed	Phase II supplement-backed quantitative branch
D12	Charged-lepton/Koide, texture, dark-sector, baryogenesis, black-hole spectroscopy, and string/worldsheet continuations	various subsets of D6, D9, D10, and D11	branch-specific EFT and phenomenological manipulations	additional ansätze such as the uniform $\mathbb{Z}_6$ center-label ensemble, discrete texture choices, dark-sector response assumptions, or discrete-horizon assumptions	Phase III phenomenological continuation / program-level branch

Accordingly, the recovered-core structural program of the compact note is D1–D5 together with D7–D9. D6 is the separate Phase-II input-dependent corollary; D10–D11 are the current Phase-II supplement-backed numerical implementation; and D12 is the Phase-III continuation layer. The only external continuous inputs are N_{scr} in D6 and P in D10, together with descendants built from those branches.

3 Five Axioms, Two External Continuous Inputs in the Current Quantitative Implementation, and Technical Premises

This section is the authoritative premise ledger and dependency map for the compact note. The OPH basis used below is exactly the five core axioms, the two external continuous inputs of the current quantitative implementation, and the theorem-local technical premises collected afterward. Older A1–A4 / B / MX / LR / R0–R1 packaging is retained elsewhere only as shorthand for pieces of this canonical package, not as a competing axiom basis. In the present version, the local finite-constraint MaxEnt/refinement branch already internalizes the quasi-local propagation and refinement-stability control that older packaging sometimes carried as separate regularity language, and Theorem 6.1 fixes the cap generator and its 2π normalization on the OPH geometric branch rather than leaving cap isotropy or Euclidean regularity as separate free premises. What remains genuinely theorem-external is the scaling-limit scope clause, the geometric-branch selection burden, the null half-sided inclusion step, fixed-cap stationarity, the bosonic/fiber-functor fork, and the

vanishing obstruction premise where invoked.

The formulation uses two external continuous configuration inputs in the current quantitative implementation:

$$P \equiv a_{\text{cell}}/\ell_P^2 = 1.63094, \tag{1}$$

$$N_{\text{scr}} \equiv \log \dim \mathcal{H}_{\text{tot}} \sim 10^{122}. \tag{2}$$

In the current implementation, P is calibrated from the gauge sector and N_{scr} is inferred from the observed cosmological constant. The first sets the particle-physics scale. The second enters the cosmological-capacity branch and a separate order-of-magnitude neutrino estimate.

Axiom 3.1 (Screen Net)**.** *Physical data is organized on a horizon screen S^2 carrying a net of local algebras*

$$P \longmapsto \mathcal{A}(P)$$

for connected patches $P \subset S^2$, with isotony

$$P \subset Q \implies \mathcal{A}(P) \subset \mathcal{A}(Q).$$

Axiom 3.2 (Overlap Consistency)**.** *For overlapping patches $P_1 \cap P_2 \neq \varnothing$, the local states induced on the shared algebra agree:*

$$\omega_{P_1}|_{\mathcal{A}(P_1 \cap P_2)} = \omega_{P_2}|_{\mathcal{A}(P_1 \cap P_2)}.$$

Axiom 3.3 (Local MaxEnt and Refinement Stability)**.** *At the regulator scale ℓ_{UV}, the realized branch is selected by maximizing entropy subject to a finite family of gauge-invariant local constraints*

$$\mathcal{C}_{\ell_{\text{UV}}} = \{O_a(x)\}_{a=1}^{N_{\text{con}}},$$

where each $O_a(x)$ is supported in a ball of radius $O(\ell_{\text{UV}})$ and the label set a is independent of the number of regulator cells. Under refinement, the same finite constraint family is preserved, so the realized states at different cutoffs belong to one common finite-dimensional MaxEnt family rather than unrelated maximizers. The realized low-energy branch is the refinement-stable branch of that family; therefore symmetry-allowed relevant operators are not held at zero by unexplained fine tuning, but only by symmetry or because they lie in the explicitly retained constraint family.

Derived local-Gibbs branch. On the finite regulator realization of Axiom 3.1, the standard finite-dimensional Lagrange-multiplier argument applied to Axiom 3.3 gives

$$\omega_{\ell_{\text{UV}}}(\lambda) = Z_{\ell_{\text{UV}}}(\lambda)^{-1} \exp\bigl(-K_{\ell_{\text{UV}}}(\lambda)\bigr),$$

with

$$K_{\ell_{\text{UV}}}(\lambda) = \sum_x \sum_{a=1}^{N_{\text{con}}} \lambda_a O_a(x) + \sum_b \mu_b Q_b,$$

where the Q_b are the finitely many optional global conserved-charge constraints. Thus the logarithm of the selected state is itself a quasi-local UV generator built from the same bounded-support densities that define the branch.

Derived quasi-local propagation and interval control. Let $\tau_t^{\ell_{\rm UV}}$ denote the automorphism group generated by $K_{\ell_{\rm UV}}(\lambda)$, or more generally by any branch generator lying in the norm-closed algebra generated by the same bounded-support local densities. Standard Lieb–Robinson estimates for finite-range interactions on the finite regulator net then give constants $C, \xi, v_{\ell_{\rm UV}} < \infty$ such that for local observables A_X, B_Y,

$$\|[\tau_t^{\ell_{\rm UV}}(A_X), B_Y]\| \le C\,\|A_X\|\,\|B_Y\|\,\min(|X|,|Y|)\,e^{-(d(X,Y)-v_{\ell_{\rm UV}}|t|)/\xi}.$$

So finite-velocity quasi-local propagation is branch-internal: it is the local finite-constraint MaxEnt branch written in propagation form, not an extra selector. The same locality statement gives the bounded-interval endpoint control later used in the null modular chain: after removal of the central endpoint term, changing an interval I to I' alters only an $O(|I'\Delta I|)$ collar of local terms. Hence on any fixed local-energy-bounded domain one has

$$\big|\langle\psi, (K[I'] - K[I])\phi\rangle\big| \le C_{\psi,\phi,I_{\max}}\,|I'\Delta I|,$$

which is precisely the endpoint-Lipschitz / finite-variation control later denoted by **T2**–**T3**. The compact note does *not* separately derive a second independent microscopic generator beyond this quasi-local branch generator; if a UV completion carries one, the only later requirement is that it lie in the same bounded-support algebraic closure so that the same propagation constants control the selected branch.

Internal refinement notion. Because Axiom 3.3 preserves the same finite local constraint family under coarse-graining, the regulator-scale states are parameterized by one common finite-dimensional multiplier space λ. The refinement-stable OPH branch is therefore the fixed point or stable manifold selected inside that multiplier space, not an imported theorem-external regularity package. Any transportable edge sector that persists along this branch defines a canonical directed system under refinement. Accordingly, whenever later sections speak of a "refinement-stable directed colimit" of edge sectors, the refinement-stable qualifier means persistence along the Axiom 3.3 branch itself; only the bosonic/symmetric/fiber-functor clauses used in gauge reconstruction remain additional categorical premises.

Axiom 3.4 (Recoverable Generalized Entropy)**.** *A generalized entropy functional exists on caps,*

$$S_{\rm gen}(C) = S_{\rm bulk}(C) + \langle L_C\rangle,$$

where L_C is a positive edge-center entropy functional. In the semiclassical scaling branch, its leading coarse-grained contribution is identified with $A(\partial C)/(4G)$. The functional obeys quantum focusing on null generators, and collar tripartitions have small conditional mutual information with controlled recovery maps [3, 4, 5, 6].

Axiom 3.5 (Minimal Admissible Realization)**.** *Among realized sector packages $\mathfrak{S}$ consisting of the connected Lie gauge-sector image relevant in the low-energy EFT, its admissible light chiral matter content, and one Higgs doublet, and which are loop-coherent, anomaly-free, refinement-stable with light chiral matter, single-Higgs Yukawa-completable with one connected abelian charge factor acting nontrivially on the coupled carrier, intrinsically CP-capable, and weak-sector UV-completable, the realized package is the lexicographically minimal one under*

$$C(\mathfrak{S}) = \big(\chi_{\rm cpl},\, N_{\rm nonab},\, N_c,\, N_g\big).$$

$\mathfrak{S}$ is the sector package on which MAR acts; it is not the bare tensor category alone. MAR is therefore an explicit structural-economy axiom on already-admissible realized low-energy branches, not a theorem derived from the preceding axioms. χ_{cpl} is the coupled carrier dimension: the smallest unitary carrier containing a common irreducible block on which the admissible pseudoreal and complex nonabelian charge types both act nontrivially. This is stronger than the abstract minimal faithful representation dimension.

Assumption 3.6 (Technical Premises)**.** *The continuum reductions still use several historical shorthand labels, but not all of them are extra premises, and the retained numbering is intentionally nonconsecutive for compatibility with earlier drafts and review surfaces. The labels **T2**–**T3** now refer only to the branch-internal quasi-local propagation and bounded-interval endpoint control already derived from Axiom 3.3; likewise, whenever later sections speak of a "refinement-stable" sector colimit, that qualifier means persistence along the same Axiom 3.3 branch rather than an additional categorical regularity axiom. The genuinely theorem-external items retained below are therefore nine in number:*

- ***T1*** *type-I local algebras at the regulator scale and boundary gauge invariance on collars;*
- ***T4*** *the branch condition that the refinement-stable cap net lies in the geometric modular class for caps and their conformal images, so that Theorem 6.1 can identify the limiting modular flow with the unique cap-preserving conformal dilation;*
- ***T6*** *a standard half-sided modular inclusion premise for the null half-line blow-up algebras;*
- ***T7*** *vanishing gluing obstruction $[z] = 0$ whenever global sector transportability is invoked;*
- ***T8*** *all Lorentz/null-modular/Einstein statements are claims about a controlled refinement/scaling limit of the regulator net, not literal exactness at fixed finite cutoff;*
- ***T9*** *fixed-cap generalized-entropy stationarity for the admissible first-variation class used in the Jacobson branch;*
- ***T10*** *symmetric braiding in the $3+1D$ EFT branch;*
- ***T11*** *a bosonic Tannakian fiber-functor premise, or else an explicit super-Tannakian fork;*
- ***T12*** *whenever compact gauge reconstruction is invoked, the sector category means the directed colimit of transportable edge sectors with objectwise finite-dimensional fibers.*

Remark 3.7 (Legacy shorthand)**.** *Historical labels are read as follows in the compact note and in mirrored surfaces: A1/A2 mean Screen Net and Overlap Consistency; B together with refinement stability means Axiom 3.3; A3 means Axiom 3.4; A4 together with the old MX / collar terminology means the recoverability and collar-control ingredients explicitly cited from Axiom 3.4; LR or the later **T2**–**T3** shorthand now mean the derived quasi-local propagation and endpoint control of the local finite-constraint MaxEnt branch, not an extra premise; and R0/R1 and $[z] = 0$ remain theorem-local technical premises.*

Remark 3.8. *The five axioms fix the OPH core, and in the present packaging they already internalize the quasi-local propagation and refinement-stable notion of sector persistence used by the scaling-limit and gauge branches. On the gravity side this means the null bridge no longer imports a separate propagation or endpoint-control premise: c.10 closes the fixed-cutoff strip argument through*

*exact-or-controlled strip additivity and the weak tail generator, while the downstream half-sided inclusion step **T6** and the scaling-limit null-stress scope clause **T8** remain explicit. The cap-modular geometric action and the 2π KMS normalization are recovered in Theorem 6.1 on the OPH geometric branch; what remains genuinely theorem-external is the retained nine-item technical package: the null half-sided inclusion step **T6**, the obstruction clause **T7** when transportability is invoked, the scaling-limit scope clause **T8**, fixed-cap stationarity **T9**, and the bosonic/symmetric/fiber-functor part of **T10**–**T12**, together with **T1** and **T4** where the fixed-cutoff regulator presentation and the geometric cap-modular branch hypothesis are used. In particular, the Lorentz/null-modular/Einstein and compact-gauge statements below are still scaling-limit claims rather than fixed-cutoff matrix-algebra identities.*

Remark 3.9 (Log convention)**.** *Unless an explicit base is written, logarithms and entropies in this compact note use natural logs (nats). Base-2 logarithms are written as* $\log_2$ *and quoted in bits.*

Authoritative dependency DAG

The table below is the authoritative dependency map for the compact note. The "Immediate OPH ingredients" column records only parents internal to the OPH program. Imported standard mathematics and extra non-axiom physical premises or external inputs are stated separately rather than hidden inside the word "derived."

Node	Compact theorem labels and output	Immediate OPH ingredients	Standard mathematics used	Extra non-axiom physical premises / external inputs	Status
D1	Theorem 4.3: unique normal form and schedule independence	overlap-consistency problem on a finite patch net	Newman's lemma for terminating locally confluent rewrite systems	finite state spaces, termination, and local confluence	structural theorem
D2	Theorem 5.2, Propositions 5.6, 5.7, 5.10, Definition 5.11: collar Markov/entropy layer	Axioms 3.1–3.4	HJPW Markov structure theorem; Petz and Fawzi–Renner recovery theory	exact Markov identities require exact Markovity or an explicitly stated idealized recoverability limit; the $A/(4G)$ identification is the coarse-grained dictionary of Definition 5.11	structural lemma/proposition layer
D3	Theorem 6.1, Corollary 6.3: geometric modular flow and Lorentz branch	Axioms 3.1–3.4	Bisognano–Wichmann modular geometry; $\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1)$	T8 together with the OPH geometric-branch hypothesis packaged in Theorem 6.1	conditional scaling-limit theorem
D4	Propositions 6.4, 6.7; Corollaries 6.5, 6.9; Theorem 6.8, Lemma 6.10; Theorem 6.11: null modular bridge to T_{kk}	D2+D3 and Axioms 3.1–3.4	Borchers–Wiesbrock half-sided modular inclusion; distributional differentiation on half-lines	the theorem-local inherited-strip decomposition and exact-or-controlled Markov hypotheses of Proposition 6.4 and Corollary 6.5, plus T6 and T8 together with the relativistic null-stress identification premise; the propagation controls denoted T2–T3 are supplied internally by Axiom 3.3 on the local finite-constraint branch	conditional bridge theorem

Node	Compact theorem labels and output	Immediate OPH ingredients	Standard mathematics used	Extra non-axiom physical premises / external inputs	Status
D5	Lemma 6.16, Theorem 6.17, Corollary 6.18: Jacobson-type Einstein branch via a rest-frame relation plus conditional tensor upgrade	D3+D4 and Axioms 3.1–3.4	Jacobson small-ball first law; Lemma 6.12 for the null-to-tensor upgrade modulo a metric term	T9 fixed-cap stationarity; locally Lorentzian $d = 4$ scaling regime; small-ball constancy assumptions; all local directions/reference states for the tensor upgrade	conditional scaling-limit theorem/corollary
D6	Proposition 6.20, Corollary 6.21: capacity branch for Λ, plus the separate neutrino estimate	D5, which already leaves the $+\Lambda g_{ab}$ ambiguity	de Sitter entropy relation and dimensional analysis	external input N_{scr} and the screen-capacity identification	input-dependent corollary / estimate
D7	Theorem 7.1: compact gauge reconstruction	Axioms 3.1–3.4	Doplicher–Roberts / Tannaka reconstruction	T7 and T10–T11 together with the directed-colimit/fiber clause of T12; the refinement-stable qualifier on that colimit is supplied internally by Axiom 3.3	conditional structural theorem
D8	Lemmas 7.3–7.8, Theorem 7.9: product gauge structure up to finite quotient	D7 + Axiom 3.5	compact Lie representation classification; Schur's lemma	connected positive-dimensional Lie admissible class; one connected abelian factor; faithful action on the minimal coupled carrier; vanishing obstruction $[z] = 0$	realized-branch theorem
D9	Theorem 7.10, Corollaries 7.11, 7.12, Proposition 7.13: realized Standard Model quotient, hypercharges, $N_c = 3$, $N_g = 3$, and product-group corollaries	D8	anomaly-cancellation algebra; Witten global-anomaly argument; CKM CP counting	realized one-generation chiral matter plus one Higgs package; asymptotic-freedom and MAR admissibility premises	realized-branch theorem/corollary chain
D10	quantitative gauge/electroweak closure of Section 7	D9 + external input P	RG evolution and matching conventions supplied in the supplement	pixel constraint; edge heat-kernel closure; supplement-backed beta-function and threshold conventions	calibration sector
D11	Higgs/top subsection of Section 7	D10	RG transport and pole-mass conversion supplied in the supplement	critical-surface condition $\lambda = \beta_\lambda = 0$ at M_U; no new continuous fit once D10 is fixed	supplement-backed quantitative branch
D12	Sections 8–10 program continuations	various subsets of D6, D9, D10, and D11	branch-specific EFT and phenomenological manipulations	additional ansätze such as the uniform $\mathbb{Z}_6$ center-label ensemble, texture choices, dark-sector response assumptions, or discrete-horizon assumptions	phenomenological continuation / program-level branch

No theorem in the compact note is meant to hide standard mathematical imports or extra physical premises inside the phrase "from the axioms." The authoritative unpacking of every such claim is the dependency DAG above. In particular, quasi-local propagation and the refinement-stable sector-persistence notion are now internal to the local finite-constraint MaxEnt branch, the cap generator and its 2π normalization are packaged inside Theorem 6.1 on the OPH geometric branch, and the remaining null half-sided inclusion, fixed-cap stationarity, and bosonic/fiber-functor

hypotheses remain explicit. The only rows that use the two external continuous inputs are D6, D10, the separate order-of-magnitude neutrino estimate attached to D6, and downstream branches built from them.

Theorem 3.10 (Summary theorem for the recovered relativity-plus-Standard-Model core)**.** *This theorem packages the recovered-core nodes D1–D5 and D7–D9 of the authoritative dependency DAG above. Node D6 is a separate input-dependent corollary, while D10–D12 remain outside this theorem's scope. Assume Axioms 3.1–3.4, Axiom 3.5, Assumption 3.6, and the hypotheses of Theorems 4.3, 6.1, 6.11, 6.17, 7.1, 7.9, and 7.10, together with Corollaries 7.11, 7.12, and Proposition 7.13. Then, in the controlled refinement/scaling regime singled out by Assumption 3.6:*

(i) overlap repair admits a unique schedule-independent normal form;

(ii) cap modular flow is geometric in the assumed modular phase and yields the connected Lorentz group

$$\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1);$$

(iii) admissible fixed-cap stationarity together with the null modular bridge and null-stress premise yield the Jacobson-type rest-frame relation

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\, \delta\langle T_{00}\rangle$$

at the cap center in the diamond rest frame; if that rest-frame relation holds for all local directions and reference states in the scaling regime, then the null-to-tensor upgrade gives

$$G_{ab} + \Lambda g_{ab} = 8\pi G\, \langle T_{ab}\rangle;$$

(iv) the refinement-stable transportable edge-sector category reconstructs a compact gauge group, and the realized connected gauge structure has the form

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\Gamma}$$

for some finite central subgroup Γ*.*

After the hypercharge lattice, generation-count, color-count, and trivial-action quotient steps, the finite quotient is fixed to $\Gamma = \mathbb{Z}_6$*, so the realized gauge structure on the realized MAR-admissible branch is*

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_c = 3, \qquad N_g = 3,$$

with the exact Standard Model hypercharge lattice on the realized matter package.

Proof. Items (i)–(iii) are collected from Theorem 4.3, Corollary 6.3, Theorem 6.17, and Corollary 6.18. The compact gauge reconstruction is Theorem 7.1; the product gauge structure up to finite quotient is Theorem 7.9; and the final identification of the exact Standard Model quotient is Theorem 7.10 together with Corollaries 7.11, 7.12, and Proposition 7.13. □

Theorem 3.10 therefore packages only the recovered relativity-plus-Standard-Model core: D1–D5 together with D7–D9. The D6 capacity relation is a separate input-dependent corollary; the quantitative calibration sector, Higgs/top branch, and phenomenological continuations remain outside its scope.

Usually Postulated Elsewhere, Recovered or Continued Here with Explicit Status

Structure	Common treatment	OPH treatment
Lorentz kinematics	background symmetry or starting axiom	conditional scaling-limit branch from geometric modular flow on screen caps
Einstein dynamics	fundamental field equation	conditional first-variation relation with tensor upgrade from generalized entropy, null modular data, and entanglement equilibrium
Gauge group	model input	conditional compact group reconstruction from edge sectors; exact SM quotient selected on the realized MAR-admissible branch
Hypercharge lattice	matter-assignment input	solved from anomaly cancellation and Yukawa invariance on the realized one-generation chiral matter plus one-Higgs package
Color and generation count	empirical input	fixed on the realized MAR-admissible branch by anomaly, CP capability, asymptotic freedom, and minimality
Flavor hierarchy unit	model-dependent small parameter	phenomenological ansatz $\varepsilon = 1/6$ motivated by a uniform $\mathbb{Z}_6$ center-label ensemble
Koide phase	phenomenological fit	continuation ansatz tied to the previously fixed discrete data as $\delta = 2/9$
Cosmological constant	local vacuum-energy puzzle	global screen-capacity term after the input-dependent capacity identification

4 Observer-Overlap Consistency as a Fixed-Point Problem

The discrete overlap problem is the mathematical skeleton of the framework.

Definition 4.1 (Patch Net)**.** *Let $G = (V, E)$ be a finite connected graph. Each vertex $i \in V$ carries a finite local state space S_i. For each edge $e = \{i, j\}$ let I_e be an interface alphabet and let*

$$\pi_{i,e} : S_i \to I_e, \qquad \pi_{j,e} : S_j \to I_e$$

be interface projections. The global state space is

$$\Sigma = \prod_{i \in V} S_i,$$

and the consistency set is

$$C = \{s \in \Sigma : \pi_{i,e}(s_i) = \pi_{j,e}(s_j) \text{ for all } e = \{i, j\} \in E\}.$$

This finiteness is a regulator-level assumption for the discrete patch-net normal-form theorem; it is not the continuum limit statement.

Define the inconsistency potential

$$\Phi(s) = \sum_{e=\{i,j\} \in E} w_e \, d_e\big(\pi_{i,e}(s_i), \pi_{j,e}(s_j)\big),$$

with $w_e > 0$ and $d_e(a, b) = 0$ iff $a = b$.

Definition 4.2 (Local Repair Law)**.** *A law λ is a family of local repair maps*

$$T_i^\lambda : \Sigma \to \Sigma$$

changing only patch i or a bounded neighborhood of i.

Theorem 4.3 (Asynchronous Fixed-Point Law)**.** *Assume:*

(i) normal forms of the repair relation are exactly the globally consistent states C;

(ii) the repair relation terminates;

(iii) the repair relation is locally confluent.

Then every $s \in \Sigma$ has a unique normal form

$$\mathrm{nf}_\lambda(s) \in C,$$

and every maximal asynchronous repair sequence terminates at that same state. A sufficient condition for (ii) is that each S_i is finite and every enabled repair strictly decreases Φ.

Proof. Termination is assumed. Local confluence plus termination imply confluence by Newman's lemma. The resulting normal form is unique. By assumption it is a globally consistent state. □

Corollary 4.4 (Schedule Independence)**.** *If observables are evaluated on $\mathrm{nf}_\lambda(s)$, the resulting physical law is independent of update order.*

Theorem 4.5 (Cycle Obstruction / Holonomy Criterion)**.** *Let A be an abelian group. For each oriented edge $e : u \to v$ assign $b_e \in A$ and consider the affine overlap equations*

$$x_v - x_u = b_e.$$

A global solution exists if and only if the signed sum of b_e vanishes on every cycle.

Proof. Summing the edge equations around a cycle gives necessity. For sufficiency, fix a root and define each x_v by summing labels along a path from the root to v. Vanishing cycle sums make this independent of the chosen path. □

Proposition 4.6 (Gauge Quotient)**.** *Suppose a local gauge group $\Gamma = \prod_i \Gamma_i$ acts on Σ while leaving all interface data invariant, and the repair maps satisfy*

$$T_i^\lambda(\gamma \cdot s) = \gamma \cdot T_i^\lambda(s).$$

Then

$$\mathrm{nf}_\lambda(\gamma \cdot s) = \gamma \cdot \mathrm{nf}_\lambda(s).$$

Hence gauge-invariant observables are unique on the quotient.

Proof. Gauge covariance maps repair sequences to repair sequences. Since the normal form is unique by Theorem 4.3, the normal form of $\gamma \cdot s$ must be $\gamma \cdot \mathrm{nf}_\lambda(s)$. □

These three results already encode much of the eventual physical interpretation. Objectivity becomes confluence, gauge symmetry becomes quotient invariance, and stable defects are represented by nontrivial overlap holonomy classes.

5 Collars, Edge Centers, and Generalized Entropy

The fixed-point picture explains why overlap consistency produces gauge quotients, but the gravity and consensus arguments later use a more specific collar fact: after edge-center completion, interior observables become insensitive to compatible exterior substitutions, and modular additivity becomes exact in the Markov normal form. The point of this section is therefore twofold. First, at fixed regulator scale and on the ordinary or central-defect branch, we derive the collar block decomposition from overlap consistency itself. Second, we state precisely when the approximate recoverability premise of Axiom 3.4 is allowed to converge to the exact HJPW normal form used by the later spatial and null collar theorems.

The sharp claim boundary is as follows. Small conditional mutual information always gives a constructive recovered comparison state with $O(\varepsilon^{1/2})$ observable error. It does *not* by itself give a universal one-shot trace-norm bound to an exact Markov state. The exact Markov normal form is recovered either when $I(A : D \mid B) = 0$ holds literally, or for a controlled family on one fixed finite-dimensional collar model, or after pullback to such a model, where $\varepsilon \to 0$. On that fixed collar one gets a collar-dependent modulus $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \to 0$ measuring distance to the exact Markov set, and this is the error that must be carried into later exact splice or modular-additivity identities.

Proposition 5.1 (Derived regulator gluing datum)**.** *Fix a regulator-scale finite patch cover of a cap neighborhood and let R be any finite union of regulator cells. Assume only the finite patch-net presentation implicit in Definition 4.1 together with overlap consistency on common interfaces. Then:*

1. *each regulator cell i with finite local state space S_i can be Hilbertized as $\tilde{\mathcal{H}}_i \cong \mathbb{C}^{|S_i|}$, so the extended algebra before overlap quotienting is the finite type-I algebra*

$$\tilde{\mathcal{A}}(R) = \mathcal{B}(\tilde{\mathcal{H}}_R), \qquad \tilde{\mathcal{H}}_R := \bigotimes_{i \subset R} \tilde{\mathcal{H}}_i;$$

2. *for each connected boundary component $\Sigma \subset \partial R$, after choosing a reference cut presentation $\tilde{\mathcal{H}}_\Sigma$, every overlap-consistent change of local chart along Σ is implemented by a unitary on $\tilde{\mathcal{H}}_\Sigma$, and the compact closure of the subgroup generated by all such recharting unitaries is a compact boundary redundancy group*

$$K_\Sigma \subset U(\tilde{\mathcal{H}}_\Sigma);$$

   before the choice of reference chart, the same data form a compact unitary groupoid of overlap-preserving transitions;

3. *if triple-overlap defects are central, the projective composition law of item 2 lifts to an honest action of a compact central extension $\widehat{K}_\Sigma$; a genuinely noncentral defect is the only obstruction to reducing the transition system to an ordinary compact group action;*

4. *in the invariant-state realization of the quotient, physical observables are the chart-independent endomorphisms, equivalently the fixed-point algebra*

$$\mathcal{A}(R) = \tilde{\mathcal{A}}(R)^{\widehat{K}_{\partial R}}, \qquad \widehat{K}_{\partial R} := \prod_{\Sigma \subset \partial R} \widehat{K}_\Sigma,$$

   with the convention $\widehat{K}_\Sigma = K_\Sigma$ whenever no central extension is needed.

Historical labels $R0$ and $R1$ are shorthand for items 1 and 4 on that branch; they are not extra axioms beyond the finite regulator presentation plus overlap redundancy.

Proof sketch. Item 1 is the Hilbertization of finite sets. For item 2, each overlap-consistent recharting acts on a finite-dimensional matrix algebra, and every *-automorphism of such an algebra is inner. After transporting all local overlap presentations to a chosen reference chart, each recharting is therefore implemented by a unitary on the cut Hilbert space. The subgroup generated by those unitaries has compact closure inside a finite-dimensional unitary group, which yields K_Σ; without fixing the reference chart one has the corresponding compact unitary groupoid. Item 3 is the usual projective-versus-central-extension lift for a central 2-cocycle. Item 4 is the quotient statement: physical observables are precisely the endomorphisms insensitive to recharting of the boundary data, hence the fixed points of the derived boundary action. If the defect is genuinely noncentral, the quotient is still well defined, but the correct bookkeeping object is the crossed-module or higher-gauge data rather than an ordinary compact group. □

Theorem 5.2 (Derived edge-center collar decomposition and exact Markov normal form)**.** *Let A-B-D be a collar tripartition realized at fixed regulator scale on the ordinary or central-defect branch of Proposition 5.1. Write $B = B_L \cup B_R$ with common interface $\Sigma = \partial C$, and let $\widehat{K}_\Sigma$ be the derived compact boundary action attached to that cut, with $\widehat{K}_\Sigma = K_\Sigma$ on the ordinary branch. In the invariant-state realization,*

$$\mathcal{H}_B = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{\widehat{K}_\Sigma}.$$

Decompose the left boundary data as

$$\tilde{\mathcal{H}}_{B_L} \cong \bigoplus_\alpha W_\alpha \otimes \mathcal{H}_{b_L^\alpha},$$

with W_α irreducible $\widehat{K}_\Sigma$-modules. Because the right half-collar carries the inverse overlap transport across the same cut, it decomposes contragrediently:

$$\tilde{\mathcal{H}}_{B_R} \cong \bigoplus_\beta W_\beta^* \otimes \mathcal{H}_{b_R^\beta}.$$

Then

$$\mathcal{H}_B \cong \bigoplus_\alpha \mathcal{H}_{b_L^\alpha} \otimes \mathcal{H}_{b_R^\alpha}, \qquad Z(\mathcal{A}(B)) = \bigoplus_\alpha \mathbb{C}\,\mathbf{1}_\alpha.$$

If, in addition, the reference state on $A \cup B \cup D$ is exact Markov,

$$I(A : D|B)_\rho = 0,$$

then the state decomposes as

$$\rho_{ABD} = \bigoplus_\alpha p_\alpha\, \rho^{(\alpha)}_{Ab_L^\alpha} \otimes \rho^{(\alpha)}_{b_R^\alpha D}.$$

The same formula holds in any limit state that is exact Markov on this fixed collar model. Thus the collar-center block decomposition is forced by overlap consistency plus the derived regulator package on the ordinary or central-defect branch; exact Markov factorization is an additional state condition, not a consequence of the block decomposition alone.

Proof. By Proposition 5.1, one may realize the physical collar states as the $\widehat{K}_\Sigma$-invariant subspace of $\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R}$. Complete reducibility of finite-dimensional unitary representations gives the displayed

decomposition of $\tilde{\mathcal{H}}_{B_L}$, and the right half-collar carries the inverse transport law across the same cut, hence the contragredient decomposition of $\tilde{\mathcal{H}}_{B_R}$. Therefore

$$\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R} \cong \bigoplus_{\alpha,\beta} (W_\alpha \otimes W_\beta^*) \otimes (\mathcal{H}_{b_L^\alpha} \otimes \mathcal{H}_{b_R^\beta}).$$

Taking $\widehat{K}_\Sigma$-invariants gives

$$\mathcal{H}_B \cong \bigoplus_{\alpha,\beta} (W_\alpha \otimes W_\beta^*)^{\widehat{K}_\Sigma} \otimes (\mathcal{H}_{b_L^\alpha} \otimes \mathcal{H}_{b_R^\beta}).$$

By Schur's lemma,

$$(W_\alpha \otimes W_\beta^*)^{\widehat{K}_\Sigma} \cong \begin{cases} \mathbb{C}, & \alpha = \beta, \\ 0, & \alpha \neq \beta, \end{cases}$$

which yields the displayed block decomposition and shows that the center is generated by the block projectors. Within each block, observables from $A \cup B$ act only on the left factor and observables from $B \cup D$ act only on the right factor. If the state is exact Markov, the standard HJPW structure theorem gives the blockwise factorized normal form [7]. The same formula is used in the idealized recoverability limit when exact identities are taken literally. □

Remark 5.3 (Status of the regulator package and the precise Markov boundary)**.** *Proposition 5.1 and Theorem 5.2 show that, on the ordinary or central-defect branch, edge-center completion is a theorem of overlap consistency plus the finite regulator presentation; no preferred quantum-link realization is being assumed there. Assumption 3.6**T1** is therefore theorem-local shorthand for this derived regulator package. The only remaining fixed-cutoff obstruction to a fully general ordinary-group theorem is a genuinely noncentral 2-group defect. The distinct issue addressed next is* state *control: exact splice and modular-additivity identities require either literal exact Markovity or a controlled family that converges to the exact Markov set on one fixed finite-dimensional collar.*

Definition 5.4 (Exact Markov set and collar-distance modulus)**.** *Fix the finite-dimensional collar Hilbert space of Theorem 5.2. Let*

$$\mathfrak{M}_{A:B:D} := \{\sigma_{ABD} : \ I(A:D \mid B)_\sigma = 0\}$$

be the set of exact Markov states on that fixed collar. Equivalently, by HJPW and Theorem 5.2, $\sigma \in \mathfrak{M}_{A:B:D}$ iff

$$\sigma_{ABD} = \bigoplus_\alpha q_\alpha \, \sigma^{(\alpha)}_{Ab_L^\alpha} \otimes \sigma^{(\alpha)}_{b_R^\alpha D}$$

for the fixed edge-center decomposition of B.

For a general state ρ_{ABD}, define its distance to the exact Markov set by

$$d_{\mathrm{M}}(\rho) := \inf_{\sigma \in \mathfrak{M}_{A:B:D}} \|\rho - \sigma\|_1,$$

and the fixed-collar exact-Markov modulus by

$$\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) := \sup\{d_{\mathrm{M}}(\rho) : I(A:D \mid B)_\rho \leq \varepsilon\}.$$

Proposition 5.5 (Controlled exact Markov-collar limit)**.** *On a fixed finite-dimensional collar, the exact Markov set $\mathfrak{M}_{A:B:D}$ is compact and*

$$\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \longrightarrow 0 \qquad \text{as } \varepsilon \downarrow 0.$$

Consequently, if $\rho^{(n)}_{ABD}$ is any sequence of states on that same collar with

$$I(A:D\mid B)_{\rho^{(n)}} \le \varepsilon_n, \qquad \varepsilon_n \to 0,$$

then there exist exact Markov states $\sigma^{(n)} \in \mathfrak{M}_{A:B:D}$ such that

$$\|\rho^{(n)} - \sigma^{(n)}\|_1 \le \delta^{\mathrm{M}}_{A:B:D}(\varepsilon_n) \longrightarrow 0.$$

This is the precise sense in which approximate recoverability converges to the exact HJPW normal form used later.

Proof. The full state space on a fixed finite-dimensional Hilbert space is compact in trace norm. Conditional mutual information is continuous there because von Neumann entropy is continuous in finite dimension. Hence $\mathfrak{M}_{A:B:D}$ is closed, so it is compact as a closed subset of a compact space.

Suppose $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \not\to 0$. Then there exist $\eta > 0$, a sequence $\varepsilon_n \downarrow 0$, and states $\rho^{(n)}$ with $I(A:D\mid B)_{\rho^{(n)}} \le \varepsilon_n$ but $d_{\mathrm{M}}(\rho^{(n)}) \ge \eta$ for all n. By compactness, pass to a trace-norm convergent subsequence $\rho^{(n_k)} \to \rho^*$. Continuity of conditional mutual information gives

$$I(A:D\mid B)_{\rho^*} = \lim_{k\to\infty} I(A:D\mid B)_{\rho^{(n_k)}} = 0,$$

so $\rho^* \in \mathfrak{M}_{A:B:D}$. But then

$$d_{\mathrm{M}}(\rho^{(n_k)}) \le \|\rho^{(n_k)} - \rho^*\|_1 \longrightarrow 0,$$

contradicting $d_{\mathrm{M}}(\rho^{(n_k)}) \ge \eta$. Thus $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \to 0$. The final statement follows by choosing $\sigma^{(n)} \in \mathfrak{M}_{A:B:D}$ within $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon_n) + 1/n$ of $\rho^{(n)}$. □

Proposition 5.6 (Approximate collar recovery)**.** *Let A-B-D be a collar tripartition satisfying*

$$I(A:D|B)_\rho \le \varepsilon,$$

and let $\mathcal{R}_{B\to BD}$ be a recovery map such that

$$F(\rho_{ABD}, (\mathrm{id}_A \otimes \mathcal{R}_{B\to BD})(\rho_{AB})) \ge e^{-\varepsilon/2}.$$

Define the recovered comparison state

$$\rho^{\mathrm{rec}}_{ABD} := (\mathrm{id}_A \otimes \mathcal{R}_{B\to BD})(\rho_{AB}), \qquad r_{\mathrm{FR}}(\varepsilon) := 2\sqrt{1-e^{-\varepsilon}} \le 2\sqrt{\varepsilon}.$$

Then

$$\|\rho_{ABD} - \rho^{\mathrm{rec}}_{ABD}\|_1 \le r_{\mathrm{FR}}(\varepsilon).$$

Consequently, for every CPTP map $\Lambda_{BD\to B'D'}$ and every bounded observable X supported on $A\cup B'$,

$$|\mathrm{Tr}[X(\mathrm{id}_A \otimes \Lambda)(\rho_{ABD} - \rho^{\mathrm{rec}}_{ABD})]| \le \|X\|_\infty \, \|(\mathrm{id}_A \otimes \Lambda)(\rho_{ABD} - \rho^{\mathrm{rec}}_{ABD})\|_1 \le \|X\|_\infty \, r_{\mathrm{FR}}(\varepsilon).$$

Proof. The fidelity lower bound and the Fuchs–van de Graaf inequality give

$$\|\rho_{ABD} - \rho^{\mathrm{rec}}_{ABD}\|_1 \le 2\sqrt{1 - F(\rho_{ABD}, \rho^{\mathrm{rec}}_{ABD})^2} \le 2\sqrt{1-e^{-\varepsilon}} = r_{\mathrm{FR}}(\varepsilon).$$

The estimate $r_{\mathrm{FR}}(\varepsilon) \le 2\sqrt{\varepsilon}$ follows from $1 - e^{-x} \le x$. Contractivity of trace norm under CPTP maps gives the transported bound, and the observable estimate is the duality between trace norm and operator norm [6]. The bound is constructive and dimension-free. What it controls is closeness to a recovered comparison state; no claim is made that $\rho^{\mathrm{rec}}_{ABD}$ is itself exact Markov. □

Proposition 5.7 (Exact splice theorem and controlled collar approximation)**.** *Under Theorem 5.2, let $\sigma_{ABD} \in \mathfrak{M}_{A:B:D}$ have exact Markov form*

$$= \sigma_{ABD} = \bigoplus_{\alpha} p_\alpha \, \sigma^{(\alpha)}_{Ab^\alpha_L} \otimes \sigma^{(\alpha)}_{b^\alpha_R D}.$$

Let $\tau^{(\alpha)}_{b^\alpha_R D'}$ be any normalized family of states compatible with the same right-boundary sectors, and define

$$\sigma'_{ABD'} = \bigoplus_{\alpha} p_\alpha \, \sigma^{(\alpha)}_{Ab^\alpha_L} \otimes \tau^{(\alpha)}_{b^\alpha_R D'}.$$

Define the common left algebra

$$\mathcal{A}_{Ab_L} := \bigoplus_{\alpha} \mathcal{B}(\mathcal{H}_{Ab^\alpha_L}) \otimes \mathbf{1}_{b^\alpha_R},$$

canonically represented on both direct-sum Hilbert spaces. Then:

1. *for every $X \in \mathcal{A}_{Ab_L}$,*
$$\mathrm{Tr}(X\sigma'_{ABD'}) = \mathrm{Tr}(X\sigma_{ABD});$$

2. *if ρ_{ABD} is any state on the same fixed collar with $I(A:D \mid B)_\rho \le \varepsilon$, and if $\sigma_\varepsilon \in \mathfrak{M}_{A:B:D}$ is chosen so that*
$$\|\rho_{ABD} - \sigma_\varepsilon\|_1 \le \delta^{\mathrm{M}}_{A:B:D}(\varepsilon),$$
then the corresponding exact splice σ'_ε satisfies
$$\left|\mathrm{Tr}(X\rho_{ABD}) - \mathrm{Tr}(X\sigma'_\varepsilon)\right| \le \|X\|_\infty \, \delta^{\mathrm{M}}_{A:B:D}(\varepsilon)$$
for all $X \in \mathcal{A}_{Ab_L}$.

Hence the exact splice identity used later is justified either at exact Markovity or along any controlled collar family for which $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon_\delta) \to 0$.

Proof. For item 1, blockwise factorization gives

$$\mathrm{Tr}(X\sigma'_{ABD'}) = \sum_{\alpha} p_\alpha \mathrm{Tr}\Big(X_\alpha \sigma^{(\alpha)}_{Ab^\alpha_L}\Big) \, \mathrm{Tr}\Big(\tau^{(\alpha)}_{b^\alpha_R D'}\Big).$$

Here X_α denotes the α-block of X in $\mathcal{A}_{Ab_L}$. Each $\tau^{(\alpha)}$ is normalized, so the second factor is 1. The same computation for σ_{ABD} gives the same value because the original right factor is likewise normalized.

For item 2, apply item 1 to σ_ε:

$$\mathrm{Tr}(X\sigma'_\varepsilon) = \mathrm{Tr}(X\sigma_\varepsilon).$$

Therefore

$$\left|\mathrm{Tr}(X\rho_{ABD}) - \mathrm{Tr}(X\sigma'_\varepsilon)\right| = \left|\mathrm{Tr}[X(\rho_{ABD} - \sigma_\varepsilon)]\right| \le \|X\|_\infty \, \|\rho_{ABD} - \sigma_\varepsilon\|_1 \le \|X\|_\infty \, \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

□

Proposition 5.8 (Transport of modular-additivity errors from the exact Markov reference)**.** *Let $K_X(\omega) := -\log \omega_X$ for a faithful reduced state on region X, and define the modular defect*

$$\Delta K(\omega) := K_{ABD}(\omega) - K_{AB}(\omega) - K_{BD}(\omega) + K_B(\omega).$$

Fix a collar and let ρ be a faithful state with $I(A : D \mid B)_\rho \le \varepsilon$. Choose $\sigma_\varepsilon \in \mathfrak{M}_{A:B:D}$ with

$$\|\rho - \sigma_\varepsilon\|_1 \le \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

Assume that the four marginals ρ_X and $(\sigma_\varepsilon)_X$ for $X \in \{ABD, AB, BD, B\}$ all satisfy

$$\rho_X \ge \lambda_* \mathbf{1}, \qquad (\sigma_\varepsilon)_X \ge \lambda_* \mathbf{1}$$

on their supports for some $\lambda_* > 0$. Then

$$\|\Delta K(\rho) - \Delta K(\sigma_\varepsilon)\|_\infty \le 4\lambda_*^{-1}\, \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

Since $\Delta K(\sigma_\varepsilon)$ is central or blockwise constant, every matrix element of the modular defect of ρ is within $4\lambda_*^{-1}\delta^{\mathrm{M}}_{A:B:D}(\varepsilon)$ of the exact Markov value.

Proof. By monotonicity of trace distance under partial trace,

$$\|\rho_X - (\sigma_\varepsilon)_X\|_1 \le \|\rho - \sigma_\varepsilon\|_1 \le \delta^{\mathrm{M}}_{A:B:D}(\varepsilon)$$

for each $X \in \{ABD, AB, BD, B\}$. On the interval $[\lambda_*, 1]$, the function $\log x$ has derivative bounded by λ_*^{-1}. By the standard integral representation of the operator logarithm, this implies the operator-norm Lipschitz bound

$$\|K_X(\rho) - K_X(\sigma_\varepsilon)\|_\infty = \|\log(\sigma_\varepsilon)_X - \log \rho_X\|_\infty \le \lambda_*^{-1}\, \|\rho_X - (\sigma_\varepsilon)_X\|_\infty \le \lambda_*^{-1}\, \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

Summing the four region contributions gives

$$\|\Delta K(\rho) - \Delta K(\sigma_\varepsilon)\|_\infty \le 4\lambda_*^{-1}\, \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

For $\sigma_\varepsilon \in \mathfrak{M}_{A:B:D}$, the exact HJPW block factorization implies that $\Delta K(\sigma_\varepsilon)$ is central or blockwise constant, so the same bound controls all matrix elements relative to the exact Markov modular-additivity value. □

Remark 5.9 (What is and is not carried forward)**.** *There are therefore two distinct quantitative controls, and the compact note keeps them separate:*

1. *the constructive one-shot recoverability bound $r_{\mathrm{FR}}(\varepsilon) = O(\varepsilon^{1/2})$, which compares the physical state to a recovered comparison state and is enough for bounded-observable statements at one regulator stage;*

2. *the fixed-collar exact-Markov modulus $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \to 0$, which compares the physical state to the exact Markov set and is the correct quantity for justifying exact splice or modular-additivity identities in a controlled limit, with an additional factor $4\lambda_*^{-1}$ when logarithms are involved.*

Later spatial and null collar theorems therefore use exact identities only in two regimes: literal exact Markovity, or a controlled refinement or scaling family in which the relevant $\delta^{\mathrm{M}}(\varepsilon_\delta)$ tends to zero after transport to one fixed finite-dimensional collar model. The manuscript does not *claim a universal dimension-free one-shot trace-norm estimate from small conditional mutual information directly to an exact Markov state.*

Proposition 5.10 (Generalized-entropy split)**.** *If the reduced cap state takes the block form*

$$\rho_C = \bigoplus_\alpha p_\alpha \left(\rho^{(\alpha)}_{\mathrm{bulk},C} \otimes \frac{\mathbf{1}^{(\alpha)}_{\mathrm{edge}}}{d_\alpha} \right),$$

then

$$S(\rho_C) = S_{\mathrm{bulk}}(C) + \mathrm{Tr}(\rho_C L_C),$$

where

$$S_{\mathrm{bulk}}(C) = H(p_\alpha) + \sum_\alpha p_\alpha S(\rho^{(\alpha)}_{\mathrm{bulk},C}), \qquad L_C = \sum_\alpha (\log d_\alpha) P_\alpha.$$

Proof. The entropy of a direct sum is

$$S\left(\bigoplus_\alpha p_\alpha \sigma_\alpha \right) = H(p_\alpha) + \sum_\alpha p_\alpha S(\sigma_\alpha).$$

Apply this identity with

$$\sigma_\alpha = \rho^{(\alpha)}_{\mathrm{bulk},C} \otimes \frac{\mathbf{1}^{(\alpha)}_{\mathrm{edge}}}{d_\alpha}.$$

Since

$$S(\sigma_\alpha) = S(\rho^{(\alpha)}_{\mathrm{bulk},C}) + \log d_\alpha,$$

the first statement follows. □

Definition 5.11 (Effective Newton constant in the refinement dictionary)**.** *In the refinement-scaling regime, if*

$$\mathrm{Tr}(\rho_C L_C) \approx N_\Sigma \bar{\ell}(t), \qquad A(\partial C) \approx N_\Sigma a_{\mathrm{cell}},$$

then matching $\mathrm{Tr}(\rho_C L_C)$ *to the area term* $A(\partial C)/(4G)$ *defines the effective Newton constant*

$$G := \frac{a_{\mathrm{cell}}}{4\bar{\ell}(t)}.$$

This is a continuum dictionary identification, not part of the exact algebraic proposition.

Proposition 5.7 is the algebraic version of interior invariance under compatible exterior substitutions. Proposition 5.8 is the precise bridge from small collar conditional mutual information to the exact modular identities later used by the spatial and null branches: at finite stage one carries a remainder, and exact identities appear only when that remainder is zero or tends to zero in the controlled collar limit. Proposition 5.10 isolates the exact entropy split, while Definition 5.11 supplies the continuum identification that turns the edge center into a gravitational coupling.

6 Modular Geometry, Relativity, and Einstein Dynamics

6.1 Geometric modular flow

The collar analysis identifies where modular nonadditivity can live, but by itself it does not determine the continuum geometric action of the limiting modular group. The correct theorem-scope is therefore branch-conditional and must be stated explicitly. What OPH supplies here is a controlled reduction from the regulator collar to the refinement limit, together with a uniqueness and normalization statement *once* the refinement-stable cap net lies in the geometric modular class. This

removes the earlier proof-sketch packaging in which cap isotropy and Euclidean regularity appeared as separate imported premises while leaving the actual remaining phase-selection burden visible.

Fix a cap $C \subset S^2$ and a shrinking collar family $(A_\delta, B_\delta, D_\delta)$ around ∂C, with $\delta \downarrow 0$ and $A_\delta \cup B_\delta \cup D_\delta$ covering a neighborhood of the cut. Write

$$\varepsilon_\delta := I(A_\delta : D_\delta \mid B_\delta)_\omega, \qquad r_{\mathrm{FR}}(\varepsilon_\delta) := 2\sqrt{1 - e^{-\varepsilon_\delta}} \le 2\sqrt{\varepsilon_\delta},$$

and, on each fixed faithful collar model,

$$\eta_\delta^{\mathrm{M}} := 4\lambda_*^{-1}\, \delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta),$$

where $\lambda_* > 0$ is the lower spectral bound used to compare modular Hamiltonians. By the collar results proved earlier, all later "exact" collar formulas are to be read in one of two ways:

1. literal exactness when the reference collar state is exact Markov; or
2. a controlled collar family for which

$$\delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta) \to 0 \qquad (\delta \downarrow 0),$$

   with the finite-stage errors $r_{\mathrm{FR}}(\varepsilon_\delta)$ and η_δ^{M} carried explicitly.

Theorem 6.1 (Geometric modular flow on caps in the OPH geometric branch)**.** *Assume Axioms 3.1–3.4, together with the fixed-cutoff regulator presentation **T1**, the branch-internal control statements denoted **T2**–**T3**, and the controlled scaling-limit scope clause **T8** of Assumption 3.6. Assume in addition that the refinement-stable MaxEnt branch selected by OPH lies in the geometric modular class for caps and their conformal images. Then for every cap $C \subset S^2$ the following hold.*

(i) *At regulator scale, the modular defect is confined to the shrinking collar up to the carried errors measured by $r_{\mathrm{FR}}(\varepsilon_\delta)$ and η_δ^{M}.*

(ii) *In the refinement limit, the modular action on $\mathcal{A}(C)$ is a one-parameter subgroup of $\mathrm{Conf}^+(S^2)$ preserving C and fixing ∂C.*

(iii) *The connected noncompact subgroup of $\mathrm{Conf}^+(S^2)$ with that property is unique; if B_C denotes its generator, then the limiting modular Hamiltonian is of the form*

$$K_C = \kappa_C\, B_C$$

for some $\kappa_C > 0$.

(iv) *The modular KMS condition fixes the normalization to*

$$\kappa_C = 2\pi.$$

Consequently, for every controlled collar family with

$$\delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta) \to 0$$

and surviving faithfulness bound, there exists a carried remainder $E_C^{(\delta)}$ such that

$$K_C^{(\delta)} = 2\pi B_C + E_C^{(\delta)},$$

where the effect of $E_C^{(\delta)}$ on bounded observables is controlled by $r_{\mathrm{FR}}(\varepsilon_\delta)$ and its modular-Hamiltonian replacement error is controlled by η_δ^{M}. In the exact-Markov case one may take $E_C^{(\delta)} = 0$ at that collar scale; in the controlled refinement limit,

$$K_C = 2\pi B_C.$$

Proof. Step 1: collar localization. At one fixed collar scale, Propositions 5.7 and 5.10 together with the controlled Markov replacement results show that all exact collar identities are justified either literally at exact Markovity or along a family whose failure of exactness is measured by $r_{\mathrm{FR}}(\varepsilon_\delta)$ and η_δ^{M}. In particular, the nonadditive part of the modular generator is confined to the shrinking collar around ∂C up to those carried errors.

Step 2: passage to the geometric branch. By the branch hypothesis, the refinement-stable cap net belongs to the geometric modular class. Hence the limiting modular automorphism group of $\mathcal{A}(C)$ acts by orientation-preserving conformal transformations of S^2 and preserves the cap C. Since the modular group also preserves the algebra of the complement, it fixes the boundary circle ∂C.

Step 3: uniqueness of the cap-preserving noncompact flow. The stabilizer in $\mathrm{Conf}^+(S^2)$ of a round circle is conjugate to the stabilizer of the equator in the Möbius group. Its connected noncompact one-parameter subgroup preserving one of the two caps is unique. Denote its infinitesimal generator by B_C. Therefore the limiting modular Hamiltonian must be proportional to B_C:

$$K_C = \kappa_C B_C$$

for some positive constant κ_C.

Step 4: normalization. The modular flow is KMS at inverse temperature 1 by definition. After analytic continuation to Euclidean modular angle, the orbit of the geometric cap-preserving flow closes with the standard angular period. Therefore the proportionality constant is fixed uniquely to

$$\kappa_C = 2\pi.$$

This is the same rigidity that fixes the standard Bisognano–Wichmann normalization; here it is stated as a property of the geometric OPH branch rather than as a separate imported Euclidean premise.

Step 5: controlled limit. At finite collar scale the only discrepancy between the physical modular generator and its exact geometric normal form is the carried collar-replacement error already isolated above. Hence one may write

$$K_C^{(\delta)} = 2\pi B_C + E_C^{(\delta)}$$

with bounded-observable and modular-Hamiltonian effects controlled respectively by $r_{\mathrm{FR}}(\varepsilon_\delta)$ and η_δ^{M}. When the collar family satisfies

$$\delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta) \to 0$$

and the lower spectral bound survives, the carried remainder vanishes in the refinement limit, yielding

$$K_C = 2\pi B_C.$$

□

Remark 6.2 (What is proved and what is not)**.** *The theorem proves geometric modular flow and its 2π normalization* on the OPH refinement branch in which the cap net has geometric modular action. *The attached source set does not prove that the bare OPH axioms alone force the existence of that branch. The remaining nontrivial burden is therefore phase selection in the refinement limit, not the collar-local reduction or the normalization once the geometric branch is realized.*

Corollary 6.3 (Lorentz Kinematics)**.** *Under the hypotheses of Theorem 6.1,*

$$\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1).$$

The cap modular flows are therefore the standard one-parameter Lorentz boost/dilation subgroups in the celestial-sphere realization, and the induced local kinematic group is the connected Lorentz group.

Proof. Orientation-preserving conformal maps of S^2 are exactly the Möbius transformations, so

$$\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1).$$

By Theorem 6.1, the cap modular groups are identified with the cap-preserving conformal dilations inside this group. Hence the local kinematics induced by modular flow is the connected Lorentz group. □

6.2 The null modular bridge

The fixed-cutoff bridge needed for the gravity chain stops one step earlier than the later stress-tensor identification. At fixed regulator scale one can transfer the same cut-center data used in the spatial collar theorem to regulated null strips, and on one inherited strip model one can obtain exact or controlled four-term strip additivity together with endpoint-Lipschitz control of the renormalized half-line family. What is not proved here is the downstream density upgrade of the weak tail generator, the half-sided modular inclusion step, the Borchers promotion to a genuine positive translation generator, or the relativistic identification of the resulting density with a continuum T_{kk}. The present subsection therefore establishes only the following c.10-local statements:

1. null-strip center transfer and the inherited left/right split hypothesis;
2. exact strip additivity on the exact inherited Markov model, and a carried defect operator on controlled exact-Markov replacements on one fixed strip model;
3. endpoint-Lipschitz matrix elements for the renormalized half-line family and the resulting weak tail generator.

The labels 6.8, 6.9, 6.10, and 6.11 are retained below only as explicit forward-reference boundaries for the downstream c.11–c.13 tasks.

Proposition 6.4 (Null-strip center transfer and inherited split)**.** *Fix a null generator Ω and a regulated tripartition*

$$I_- = (v_1, v_2), \qquad J = (v_2, v_3), \qquad I_+ = (v_3, v_4),$$

with cuts

$$\Gamma_- := \{v = v_2\}, \qquad \Gamma_+ := \{v = v_3\}.$$

*Assume the fixed-cutoff regulator package **T1** of Assumption 3.6, and assume that the two null cuts inherit the ordinary or central-defect boundary-redundancy data of Proposition 5.1 in the following precise sense:*

(i) each cut $\Gamma_\pm$ carries a compact derived boundary action $\widehat{K}_\pm$ on a reference cut Hilbert space, with $\widehat{K}_\pm = K_\pm$ on the ordinary branch;

(ii) in a compatible type-I regulator presentation, the adjacent regions carry inverse transport across each cut, so for irreducible $\widehat{K}_\pm$-modules $W_{\alpha_\pm}$ one has decompositions

$$\tilde{\mathcal{H}}_{I_-} \cong \bigoplus_{\alpha_-} W_{\alpha_-} \otimes \mathcal{H}_{i_-^{\alpha_-}},$$

$$\tilde{\mathcal{H}}_J \cong \bigoplus_{\alpha_-,\alpha_+} W^*_{\alpha_-} \otimes \mathcal{H}_{j^{\alpha_-,\alpha_+}} \otimes W_{\alpha_+},$$

$$\tilde{\mathcal{H}}_{I_+} \cong \bigoplus_{\alpha_+} W^*_{\alpha_+} \otimes \mathcal{H}_{i_+^{\alpha_+}};$$

(iii) the gauge-invariant strip algebra is the commutant of the transported boundary action on $\tilde{\mathcal{H}}_J$, so that the pair (α_-,α_+) is a central cut label;

(iv) for each (α_-,α_+), the multiplicity space $\mathcal{H}_{j^{\alpha_-,\alpha_+}}$ admits a chosen factorization

$$\mathcal{H}_{j^{\alpha_-,\alpha_+}} \cong \mathcal{H}_{j_L^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-,\alpha_+}}$$

such that, blockwise, $\mathcal{A}(I_- \cup J)$ acts only on $\mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j_L^{\alpha_-,\alpha_+}}$ and $\mathcal{A}(J \cup I_+)$ acts only on $\mathcal{H}_{j_R^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{i_+^{\alpha_+}}$.

Then the strip algebra has the inherited central decomposition

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{B}(\mathcal{H}_{j^{\alpha_-,\alpha_+}}), \qquad Z(\mathcal{A}(J)) = \bigoplus_{\alpha_-,\alpha_+} \mathbb{C}\, P_{\alpha_-,\alpha_+},$$

and, under item (iv),

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{B}(\mathcal{H}_{j_L^{\alpha_-,\alpha_+}}) \otimes \mathcal{B}(\mathcal{H}_{j_R^{\alpha_-,\alpha_+}}).$$

Moreover the glued tripartition carries the block decomposition

$$\mathcal{H}_{I_- \cup J \cup I_+} \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j_L^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{i_+^{\alpha_+}}.$$

Thus the two null cuts transfer the same center data as the spatial collar branch, while the left/right split of the strip multiplicity spaces is exactly the extra decomposition-inheritance hypothesis and is not forced by center transfer alone.

Proof. By complete reducibility of the finite-dimensional unitary actions $\widehat{K}_\pm$, the displayed decompositions of $\tilde{\mathcal{H}}_{I_-}$, $\tilde{\mathcal{H}}_J$, and $\tilde{\mathcal{H}}_{I_+}$ exist. Under item (iii), the strip algebra is the commutant of the transported $\widehat{K}_- \times \widehat{K}_+$ action on

$$\tilde{\mathcal{H}}_J = \bigoplus_{\alpha_-,\alpha_+} W^*_{\alpha_-} \otimes \mathcal{H}_{j^{\alpha_-,\alpha_+}} \otimes W_{\alpha_+}.$$

Write an operator $X \in \mathcal{B}(\tilde{\mathcal{H}}_J)$ in matrix form relative to that direct sum. The block

$$X_{(\alpha_-,\alpha_+),(\beta_-,\beta_+)}$$

intertwines

$$W^*_{\alpha_-} \otimes W_{\alpha_+} \longrightarrow W^*_{\beta_-} \otimes W_{\beta_+}$$

for the product action of $\widehat{K}_- \times \widehat{K}_+$. By Schur's lemma, such an intertwiner is zero unless $(\alpha_-, \alpha_+) = (\beta_-, \beta_+)$, and when the sector pair agrees it is scalar on the representation factors. Therefore the commutant is exactly

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{B}(\mathcal{H}_{j^{\alpha_-,\alpha_+}}),$$

and the center is generated by the corresponding block projectors P_{α_-,α_+}.

For the glued tripartition, tensor the three regulator Hilbert spaces and decompose:

$$\tilde{\mathcal{H}}_{I_-} \otimes \tilde{\mathcal{H}}_J \otimes \tilde{\mathcal{H}}_{I_+} \cong \bigoplus_{\alpha_-,\beta_-,\alpha_+,\beta_+} (W_{\alpha_-} \otimes W^*_{\beta_-}) \otimes \mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j^{\beta_-,\beta_+}} \otimes (W_{\beta_+} \otimes W^*_{\alpha_+}) \otimes \mathcal{H}_{i_+^{\alpha_+}}.$$

Taking $\widehat{K}_- \times \widehat{K}_+$-invariants and applying Schur's lemma to each cut gives

$$(W_{\alpha_-} \otimes W^*_{\beta_-})^{\widehat{K}_-} \cong \begin{cases} \mathbb{C}, & \alpha_- = \beta_-, \\ 0, & \alpha_- \neq \beta_-, \end{cases} \qquad (W_{\beta_+} \otimes W^*_{\alpha_+})^{\widehat{K}_+} \cong \begin{cases} \mathbb{C}, & \beta_+ = \alpha_+, \\ 0, & \beta_+ \neq \alpha_+, \end{cases}$$

so only matching cut sectors survive and one obtains

$$\mathcal{H}_{I_- \cup J \cup I_+} \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{i_+^{\alpha_+}}.$$

If item (iv) holds, substitute the chosen factorization

$$\mathcal{H}_{j^{\alpha_-,\alpha_+}} \cong \mathcal{H}_{j_L^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-,\alpha_+}}$$

to obtain the displayed inherited split and the stated action of the left and right union algebras. Nothing in the Schur-lemma argument itself forces item (iv); it is exactly the extra structural condition required to reproduce the same blockwise tensor pattern as Theorem 5.2. □

Corollary 6.5 (Exact or controlled four-term null modular relation on an inherited strip model)**.** *Fix one finite-dimensional regulated strip model satisfying Proposition 6.4, and define*

$$\mathfrak{M}_{I_-:J:I_+} := \{\sigma : \ I(I_- : I_+ \mid J)_\sigma = 0\},$$

$$\delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon) := \sup \left\{ \inf_{\sigma \in \mathfrak{M}_{I_-:J:I_+}} \|\rho - \sigma\|_1 : I(I_- : I_+ \mid J)_\rho \leq \varepsilon \right\}.$$

For any state η on this strip model, write

$$\Delta K_J(\eta) := K_{I_- \cup J \cup I_+}(\eta) - K_{I_- \cup J}(\eta) - K_{J \cup I_+}(\eta) + K_J(\eta).$$

If the reference state ω is exact Markov on this inherited decomposition, then

$$\Delta K_J(\omega) \in Z(\mathcal{A}(J)),$$

and in the canonical blockwise identification of Proposition 6.4 one may take

$$\Delta K_J(\omega) = 0.$$

Equivalently, without fixing that normalization there exists a central operator

$$K_{\partial,J}(\omega) \in Z(\mathcal{A}(J))$$

such that

$$K_{I_-\cup J\cup I_+}(\omega) = K_{I_-\cup J}(\omega) + K_{J\cup I_+}(\omega) - K_J(\omega) + K_{\partial,J}(\omega).$$

If instead $I(I_- : I_+ \mid J)_\omega \le \varepsilon$*, then on this fixed strip model one may choose an exact Markov state* $\widetilde{\omega}_J \in \mathfrak{M}_{I_-:J:I_+}$ *with*

$$\|\omega - \widetilde{\omega}_J\|_1 \le \delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon).$$

Define the carried defect operator

$$\mathfrak{D}_J(\omega, \widetilde{\omega}_J) := \Delta K_J(\omega) - \Delta K_J(\widetilde{\omega}_J).$$

Then

$$K_{I_-\cup J\cup I_+}(\omega) = K_{I_-\cup J}(\omega) + K_{J\cup I_+}(\omega) - K_J(\omega) + K_{\partial,J}(\widetilde{\omega}_J) + \mathfrak{D}_J(\omega, \widetilde{\omega}_J),$$

where

$$K_{\partial,J}(\widetilde{\omega}_J) := \Delta K_J(\widetilde{\omega}_J) \in Z(\mathcal{A}(J)).$$

For every bounded observable O *in* $\mathcal{A}(I_- \cup J)$ *or* $\mathcal{A}(J \cup I_+)$*,*

$$\big|\omega(O) - \widetilde{\omega}_J(O)\big| \le \|O\|_\infty\, \delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon).$$

If the marginals entering ΔK_J *for* ω *and* $\widetilde{\omega}_J$ *are uniformly faithful with lower spectral bound* $\lambda_* > 0$*, then*

$$\|\mathfrak{D}_J(\omega, \widetilde{\omega}_J)\|_\infty \le 4\lambda_*^{-1}\, \delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon).$$

Independently, the Fawzi–Renner recovery map supplies a constructive comparison state with trace-norm error

$$r_{\mathrm{FR}}(\varepsilon) = 2\sqrt{1 - e^{-\varepsilon}} \le 2\sqrt{\varepsilon}.$$

Hence the exact four-term strip relation is available only at exact Markovity, or along a controlled strip family on one fixed inherited strip model for which

$$\delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon_J) \to 0.$$

In the controlled case, $\mathfrak{D}_J$ *is carried explicitly at each finite stage.*

Proof. Assume first that ω is exact Markov on the inherited split of Proposition 6.4. Then the HJPW structure theorem gives, on each sector pair $\alpha = (\alpha_-, \alpha_+)$,

$$\omega_{I_- J I_+}|_\alpha = p_\alpha\, \omega^{(\alpha)}_{I_- j_L} \otimes \omega^{(\alpha)}_{j_R I_+}.$$

Accordingly,

$$K_{I_-\cup J\cup I_+}(\omega)\big|_\alpha = (-\log p_\alpha)\mathbf{1} + K^{(\alpha)}_{I_- j_L} \otimes \mathbf{1}_{j_R I_+} + \mathbf{1}_{I_- j_L} \otimes K^{(\alpha)}_{j_R I_+}.$$

Tracing over I_+, I_-, or both gives the marginal block formulas

$$K_{I_-\cup J}(\omega)\big|_\alpha = (-\log p_\alpha)\mathbf{1} + K^{(\alpha)}_{I_- j_L} \otimes \mathbf{1}_{j_R} + \mathbf{1}_{I_- j_L} \otimes K^{(\alpha)}_{j_R},$$

$$K_{J\cup I_+}(\omega)\big|_\alpha = (-\log p_\alpha)\mathbf{1} + K^{(\alpha)}_{j_L} \otimes \mathbf{1}_{j_R I_+} + \mathbf{1}_{j_L} \otimes K^{(\alpha)}_{j_R I_+},$$

$$K_J(\omega)\big|_\alpha = (-\log p_\alpha)\mathbf{1} + K^{(\alpha)}_{j_L} \otimes \mathbf{1}_{j_R} + \mathbf{1}_{j_L} \otimes K^{(\alpha)}_{j_R}.$$

Subtracting the last three expressions from the first gives zero on every block. Thus, in the canonical blockwise identification supplied by Proposition 6.4,

$$\Delta K_J(\omega) = 0.$$

If one chooses to keep the blockwise endpoint-label bookkeeping explicit rather than absorbing those constants into the canonical identification, the remainder is a direct sum of block constants and is therefore central. This yields the stated central form with $K_{\partial,J}(\omega) \in Z(\mathcal{A}(J))$.

For the controlled statement, apply Proposition 5.5 on this fixed finite-dimensional strip model, relabeling A, B, D there as I_-, J, I_+. This yields an exact Markov replacement $\widetilde{\omega}_J \in \mathfrak{M}_{I_-:J:I_+}$ with the displayed trace-norm bound. The identity

$$\Delta K_J(\omega) = \Delta K_J(\widetilde{\omega}_J) + \mathfrak{D}_J(\omega, \widetilde{\omega}_J)$$

is tautological, so the displayed four-term formula follows from the exact-Markov case applied to $\widetilde{\omega}_J$. The observable estimate is immediate from

$$|\omega(O) - \widetilde{\omega}_J(O)| \leq \|O\|_\infty \, \|\omega - \widetilde{\omega}_J\|_1.$$

Finally, Proposition 5.8, again relabeled $A, B, D \mapsto I_-, J, I_+$, yields the operator-norm bound on $\mathfrak{D}_J$ whenever the relevant marginals have the stated lower spectral bound. The Fawzi–Renner estimate is the same one-shot recoverability bound already used elsewhere and is independent of the exact-Markov replacement modulus. □

Definition 6.6 (Renormalized null modular functional)**.** *For a null interval I on generator Ω, let $K_\partial(I, \Omega)$ denote the central endpoint-label term singled out by Corollary 6.5 on the inherited strip model, or by its controlled exact-Markov replacement when that model is used as reference. Define*

$$\widetilde{K}[I, \Omega] := K[I, \Omega] - K_\partial(I, \Omega).$$

For a null half-line $H_a := (a, \infty)$, abbreviate

$$\widetilde{K}_a(\Omega) := \widetilde{K}[H_a, \Omega].$$

Proposition 6.7 (Endpoint-Lipschitz null modular families and weak tail generator)**.** *Under the derived quasi-local propagation and bounded-interval endpoint control already internal to Axiom 3.3, the renormalized interval family obeys the following matrix-element bounds on every fixed local-energy-bounded domain. For bounded intervals*

$$I = (a, b), \qquad I' = (a', b'),$$

contained in one compact endpoint window,

$$\left|\langle \psi, (\widetilde{K}[I', \Omega] - \widetilde{K}[I, \Omega])\phi \rangle\right| \leq C_{\psi,\phi,\Omega} \left(|a' - a| + |b' - b|\right).$$

In particular the matrix elements of $\widetilde{K}[I, \Omega]$ are jointly Lipschitz in the two endpoints and therefore have finite variation there.

For null half-lines $H_a = (a, \infty)$, one likewise has

$$\left|\langle \psi, (\widetilde{K}_{a'}(\Omega) - \widetilde{K}_a(\Omega))\phi \rangle\right| \leq C_{\psi,\phi,\Omega} \, |a' - a|$$

for a, a' in every compact window. Hence, for fixed ψ, ϕ,

$$f_{\psi,\phi}(a) := \langle \psi, \widetilde{K}_a(\Omega)\phi \rangle$$

is locally Lipschitz and therefore absolutely continuous. Its distributional derivative defines a locally L^∞ sesquilinear-form density

$$\langle \psi, q(a, \Omega)\phi \rangle := -\frac{1}{2\pi} \, \partial_a f_{\psi,\phi}(a),$$

and for $a < b$,

$$\langle \psi, (\widetilde{K}_b(\Omega) - \widetilde{K}_a(\Omega))\phi\rangle = -2\pi \int_a^b \langle \psi, q(v,\Omega)\phi\rangle \, dv.$$

The object $q(a,\Omega)$ is the weak tail generator. At this stage it is not yet asserted to be positive or self-adjointly implemented.

Proof. The bounded-interval endpoint-control statement already derived from Axiom 3.3 applies precisely to the renormalized family obtained after removal of the central endpoint term. Therefore, for intervals contained in one fixed compact endpoint window,

$$\left|\langle \psi, (\widetilde{K}[I',\Omega] - \widetilde{K}[I,\Omega])\phi\rangle\right| \le C_{\psi,\phi,\Omega}\, |I'\Delta I|,$$

where $I'\Delta I$ is the symmetric difference of the two intervals. If

$$I = (a,b), \qquad I' = (a',b'),$$

then

$$|I'\Delta I| \le |a'-a| + |b'-b|,$$

which gives the displayed joint endpoint-Lipschitz bound.

For half-lines $H_a = (a,\infty)$ and $H_{a'} = (a',\infty)$, the symmetric difference is the bounded interval between the two endpoints, so

$$|H_{a'}\Delta H_a| = |a'-a|.$$

Because the central endpoint-label term has already been removed in the definition of $\widetilde{K}_a$, the same endpoint-control estimate yields

$$\left|\langle \psi, (\widetilde{K}_{a'}(\Omega) - \widetilde{K}_a(\Omega))\phi\rangle\right| \le C_{\psi,\phi,\Omega}\, |a'-a|.$$

Hence, on every compact a-interval, the function

$$f_{\psi,\phi}(a) := \langle \psi, \widetilde{K}_a(\Omega)\phi\rangle$$

is Lipschitz. A Lipschitz function on a compact interval is absolutely continuous, so it has an almost-everywhere derivative in L^∞ and obeys the fundamental theorem of calculus:

$$f_{\psi,\phi}(b) - f_{\psi,\phi}(a) = \int_a^b \partial_v f_{\psi,\phi}(v)\, dv.$$

Define

$$\langle \psi, q(v,\Omega)\phi\rangle := -\frac{1}{2\pi}\, \partial_v f_{\psi,\phi}(v)$$

distributionally. Substituting this definition into the previous identity gives

$$\langle \psi, (\widetilde{K}_b(\Omega) - \widetilde{K}_a(\Omega))\phi\rangle = -2\pi \int_a^b \langle \psi, q(v,\Omega)\phi\rangle \, dv.$$

This is exactly the claimed weak-tail formula. □

Theorem 6.8 (Downstream density-upgrade template)**.** *Fix a null generator Ω and the renormalized half-line family $\widetilde{K}_a(\Omega)$ of Proposition 6.7. For fixed vectors ψ, ϕ in a common dense domain, let*

$$f_{\psi,\phi}(a) := \langle \psi, \widetilde{K}_a(\Omega)\phi\rangle, \qquad q_{\psi,\phi}(a) := \langle \psi, q(a,\Omega)\phi\rangle = -\frac{1}{2\pi}\, \partial_a f_{\psi,\phi}(a).$$

Assume in addition that $q_{\psi,\phi}$ is weakly differentiable in a, that

$$\lim_{a\to+\infty} f_{\psi,\phi}(a) = 0, \qquad \lim_{a\to+\infty} q_{\psi,\phi}(a) = 0,$$

and define distributionally

$$\langle \psi, p(a,\Omega)\phi\rangle := -\partial_a q_{\psi,\phi}(a) = \frac{1}{2\pi}\,\partial_a^2 f_{\psi,\phi}(a).$$

Then

$$q_{\psi,\phi}(a) = \int_a^\infty \langle \psi, p(v,\Omega)\phi\rangle\, dv,$$

and

$$\langle \psi, \widetilde{K}_a(\Omega)\phi\rangle = 2\pi \int_a^\infty (v-a)\,\langle \psi, p(v,\Omega)\phi\rangle\, dv.$$

This statement is not part of the present fixed-cutoff bridge; it records the exact additional input/output package carried into the downstream density-upgrade task.

Corollary 6.9 (Geometric form of the downstream half-sided inclusion step)**.** *Assume in addition the standard half-sided modular inclusion premise **T6** for the null half-line blow-up algebras,*

$$\sigma_t^\omega(\mathcal{A}(v>v_1)) \subseteq \mathcal{A}(v>v_1) \qquad (t\geq 0).$$

Then the blow-up limit of Theorem 6.1 near a smooth entangling cut acts geometrically by the null dilation

$$v - v_0 \mapsto e^{-2\pi t}(v-v_0),$$

*which is the geometric realization of the assumed half-sided inclusion on null half-lines. This step does not prove **T6** from fixed-cutoff strip data; it identifies the geometric form that **T6** takes on the OPH geometric branch.*

Proof. Near a smooth cut, the cap dilation from Theorem 6.1 becomes a Rindler boost in the tangent geometry. Restricting that boost to a null generator gives the stated dilation. For $t \geq 0$, the half-line $v > v_0$ maps into itself, which is exactly the geometric form of the assumed half-sided inclusion. □

Lemma 6.10 (Downstream Borchers promotion template)**.** *Under the half-sided modular inclusion premise of Corollary 6.9, the weak tail generator of Proposition 6.7 is promoted to a genuine one-parameter unitary group $U(a) = e^{iaP}$ with $P \geq 0$ such that*

$$\Delta^{it} U(a) \Delta^{-it} = U(e^{-2\pi t} a), \qquad [K,P] = i2\pi P.$$

This lemma is a downstream Borchers–Wiesbrock step and is not proved from the fixed-cutoff strip arguments above.

Theorem 6.11 (Downstream null-stress identification template)**.** *Assume Axioms 3.1–3.4, Assumption 3.6, Proposition 6.4, the exact-or-controlled strip hypothesis of Corollary 6.5, Proposition 6.7, the density-upgrade hypotheses of Theorem 6.8, the half-sided inclusion premise **T6**, and the scaling-limit null-stress identification premise of the relativistic EFT branch. Then there exists an operator-valued distribution $T_{kk}(v,\Omega)$ such that the renormalized modular Hamiltonian on a null half-line takes the form*

$$\widetilde{K}[(a,\infty),\Omega] = 2\pi \int_a^\infty (v-a)\, T_{kk}(v,\Omega)\, dv + E^{(\eta)}_{(a,\infty),\Omega},$$

where for each relevant half-line or bounded interval I, ε_I denotes the strip-CMI control parameter and δ_I^{M} the corresponding fixed-strip exact-Markov modulus of Corollary 6.5, and

$$\eta_I := r_{\mathrm{FR}}(\varepsilon_I) + 4\lambda_*^{-1}\delta_I^{\mathrm{M}}(\varepsilon_I) + \eta_I^{\mathrm{EFT}},$$

and for all domain vectors ψ, ϕ,

$$\left|\langle \psi, E_{I,\Omega}^{(\eta)}\phi\rangle\right| \le C_{I,\Omega,\psi,\phi}\,\eta_I.$$

If bounded null intervals inherit geometric modular action from the interval-preserving projective flow, then for each bounded interval I one likewise has

$$\widetilde{K}[I,\Omega] = 2\pi\int_I g_I(v)\,T_{kk}(v,\Omega)\,dv + E_{I,\Omega}^{(\eta)},$$

where $g_I(v)$ is affine-covariant and vanishes at finite interval endpoints. The full modular Hamiltonian is then

$$K[I,\Omega] = \widetilde{K}[I,\Omega] + K_\partial(I,\Omega),$$

with K_∂ central and boundary-supported. Knowledge of T_{kk} for all null directions reconstructs a symmetric tensor T_{ab} up to the addition of a metric term ϕg_{ab}. This theorem is not established at the present c.10 step; it records the exact additional c.11–c.13 burden beyond the fixed-cutoff bridge proved above.

Lemma 6.12 (Null data determine the stress tensor modulo a metric term)**.** *Let X_{ab} be a symmetric tensor. If*

$$X_{ab}k^a k^b = 0$$

for every null vector k at a point, then

$$X_{ab} = \phi g_{ab}$$

for some scalar ϕ.

Proof. Write

$$X_{ab} = Y_{ab} + \phi g_{ab}$$

with Y_{ab} traceless. Since $g_{ab}k^a k^b = 0$ on null vectors, the hypothesis implies $Y_{ab}k^a k^b = 0$ for all null k. In local inertial coordinates let $k = (1, \hat{n})$ with $|\hat{n}| = 1$. Then

$$Y_{ab}k^a k^b = Y_{00} + 2\hat{n}^i Y_{0i} + \hat{n}^i\hat{n}^j Y_{ij}.$$

Oddness under $\hat{n} \mapsto -\hat{n}$ forces $Y_{0i} = 0$. Averaging the remaining relation over S^2 gives

$$Y_{00} + \frac{1}{3}\delta^{ij}Y_{ij} = 0.$$

Tracelessness of Y_{ab} gives

$$-Y_{00} + \delta^{ij}Y_{ij} = 0,$$

so both Y_{00} and the spatial trace $\delta^{ij}Y_{ij}$ vanish. The remaining condition is therefore

$$\hat{n}^i\hat{n}^j Y_{ij}^{\mathrm{TF}} = 0 \qquad \text{for all } \hat{n} \in S^2,$$

where Y_{ij}^{TF} is the traceless spatial part. The $\ell = 2$ spherical harmonics are linearly independent, so this implies $Y_{ij}^{\mathrm{TF}} = 0$. Hence $Y_{ab} = 0$ and therefore $X_{ab} = \phi g_{ab}$. □

Remark 6.13 (Claim boundary in the null bridge)**.** *The c.10-local bridge is now closed exactly at the weak-tail-generator stage. What has been proved here is the fixed-cutoff transfer of cut-center data to null strips, the exact-or-controlled four-term strip relation on one inherited strip model, and endpoint-Lipschitz control of the renormalized half-line family strong enough to define the weak tail generator $q(a, \Omega)$. The density upgrade of q, the half-sided modular inclusion step, the Borchers promotion to a genuine positive translation generator, and the continuum null-stress identification remain explicit downstream c.11–c.13 tasks rather than hidden proof sketches.*

6.3 Generalized entropy and the Einstein equation

From this point through Corollary 6.18 we specialize to the physical $d = 4$ scaling branch.

For a reference state ω and a small cap C,

$$\delta S_C = \delta\langle K_C\rangle.$$

By Theorem 6.1,

$$\delta S_C = 2\pi\,\delta\langle B_C\rangle.$$

By Proposition 5.10,

$$S_{\text{gen}}(C) = \operatorname{Tr}(\rho L_C) + S_{\text{bulk}}(C).$$

Together with Definition 5.11, this identifies the area term as the coarse-grained form of the edge entropy density rather than an independent postulate.

Definition 6.14 (Admissible equilibrium variation)**.** *Throughout this subsection, δ denotes a first-order state variation about a reference state ω, within the cap-label-preserving MaxEnt class, at fixed cap C, fixed cap size/volume, and fixed conserved charges. Shape or null-cut deformations are denoted separately by δ_{shape} or ∂_λ.*

Assumption 6.15 (Fixed-cap generalized-entropy stationarity)**.** *For admissible variations preserving cap labels, cap size/volume, and relevant conserved charges, the realized reference state is stationary for generalized entropy:*

$$\delta S_{\text{gen}}(C) = 0.$$

This is the Jacobson-type fixed-cap stationarity premise used in the compact note. MaxEnt alone does not derive it here.

Lemma 6.16 (d=4 small-ball modular bridge)**.** *Assume the hypotheses of Theorem 6.11 and a locally Lorentzian scaling regime in which $\delta\langle T_{00}\rangle$ is approximately constant across a small radius-ℓ ball/diamond centered at the cap point. Then in $d = 4$,*

$$\delta S_C = 2\pi\,\delta\langle B_C\rangle = 2\pi \int_{B_\ell} \frac{\ell^2 - r^2}{2\ell}\,\delta\langle T_{00}\rangle\, d^3x = \frac{8\pi^2\ell^4}{15}\,\delta\langle T_{00}\rangle + O(\ell^5\partial T).$$

Proof sketch. Theorem 6.11 supplies the null modular/stress-tensor identification in the scaling branch. In the locally Lorentzian $d = 4$ small-ball limit, the cap-preserving conformal generator is the standard ball modular kernel $(\ell^2 - r^2)/(2\ell)$. Integrating that kernel over the three-ball gives

$$\int_{B_\ell} \frac{\ell^2 - r^2}{2\ell}\, d^3x = \frac{4\pi\ell^4}{15},$$

and multiplying by the overall 2π factor from the first law yields the stated coefficient. □

Theorem 6.17 (Conditional Jacobson-type Einstein relation from entanglement equilibrium)**.** *Assume Axioms 3.1–3.4, Assumption 3.6, Assumption 6.15, and the hypotheses of Theorem 6.11 and Lemma 6.16. Then, for every sufficiently small cap C and every admissible first variation δ about a reference state ω in the locally Lorentzian $d = 4$ scaling regime,*

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\, \delta\langle T_{00}\rangle$$

at the cap center in the diamond rest frame.

Proof sketch. Assumption 6.15 gives the fixed-cap stationarity condition

$$\delta S_{\mathrm{gen}}(C) = 0.$$

Lemma 6.16 identifies the bulk modular contribution in the $d = 4$ small-ball regime as

$$\delta S_{\mathrm{bulk}}(C) = \frac{8\pi^2\ell^4}{15}\, \delta\langle T_{00}\rangle.$$

For a fixed-volume small ball in $d = 4$, the standard area variation is

$$\delta A\big|_{V,\Lambda_{\mathrm{ref}}} = -\frac{4\pi\ell^4}{15}\, \delta(G_{00} + \Lambda_{\mathrm{ref}} g_{00})\,.$$

Since $S_{\mathrm{gen}} = S_{\mathrm{bulk}} + A/(4G)$ in the coarse-grained scaling dictionary, stationarity gives

$$0 = \delta S_{\mathrm{bulk}} + \frac{\delta A}{4G} = \frac{8\pi^2\ell^4}{15}\, \delta\langle T_{00}\rangle - \frac{\pi\ell^4}{15G}\, \delta(G_{00} + \Lambda_{\mathrm{ref}} g_{00})\,,$$

which rearranges to the stated rest-frame relation. □

Corollary 6.18 (Tensor Upgrade in the Scaling Regime)**.** *If the first-variation relation of Theorem 6.17 holds for all local directions and reference states in the scaling regime, then*

$$G_{ab} + \Lambda g_{ab} = 8\pi G\, \langle T_{ab}\rangle$$

modulo the metric-term ambiguity already isolated by Lemma 6.12.

Proof. Overlap consistency across observers through the same point supplies all timelike directions, and Theorem 6.11 together with Lemma 6.12 upgrades the rest-frame relation to the full tensor equation in the Jacobson entanglement-equilibrium line of argument [8, 9]. □

6.4 The cosmological constant

Lemma 6.19 (Vacuum-Energy Blindness)**.** *For any null vector k and any vacuum-energy contribution $T^{\mathrm{vac}}_{ab} = -\rho_{\mathrm{vac}} g_{ab}$,*

$$T^{\mathrm{vac}}_{kk} = 0.$$

Proposition 6.20 (Structural Separation of Λ)**.** *Local null-modular data fix T_{ab} only up to ϕg_{ab}. Therefore Λ is not determined by local overlap consistency and requires a global input.*

Corollary 6.21 (Cosmological Constant from Capacity)**.** *If the global screen capacity is identified with the de Sitter static-patch entropy,*

$$N_{\mathrm{scr}} = S_{\mathrm{dS}} = \frac{A_{\mathrm{dS}}}{4G} = \frac{3\pi}{G\Lambda},$$

then

$$\Lambda = \frac{3\pi}{G N_{\mathrm{scr}}}.$$

This is a structural aspect of the framework. The vacuum-energy problem is reorganized by separating local null data from the global capacity term.

7 Gauge Reconstruction and Standard Model Structure

7.1 Compact gauge reconstruction

At any fixed UV cutoff, edge-center completion equips collars with finitely many sector labels and fusion rules. The category used for reconstruction is the refinement-stable directed colimit

$$\mathsf{Sect}_\infty := \varinjlim_r \mathsf{Sect}_r$$

over regulator refinements r, retaining only transportable sectors and intertwiners that persist under refinement. This is the object on which Tannaka/Doplicher-Roberts reconstruction is applied.

Theorem 7.1 (Conditional compact gauge reconstruction in the bosonic branch)**.** *Assume Axioms 3.1–3.4, the relevant parts of Assumption 3.6, and that the refinement-stable edge-sector category* Sect_∞ *in the EFT regime is a rigid symmetric* C^**-tensor category with a faithful bosonic fiber functor to finite-dimensional Hilbert spaces. Then*

$$\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G)$$

for a compact group G *uniquely determined up to isomorphism.*

Proof sketch. Each fixed-cutoff category has only finitely many simple blocks, but the refinement-stable colimit Sect_∞ can have infinitely many simple objects while still carrying objectwise finite-dimensional fibers. In the bosonic EFT branch, symmetry and rigidity place the category in the Doplicher-Roberts/Tannakian reconstruction class, and the faithful bosonic fiber functor identifies it with $\mathrm{Rep}(G)$ for a compact group G [10, 11]. □

If a fermionic sign object is present, the correct reconstruction statement is super-Tannakian rather than purely Tannakian. The compact note does not prove that fermionic/chiral extension; it works only in the bosonic internal-gauge branch.

7.2 Minimal admissibility and the Standard Model quotient

Throughout this subsection, χ_{cpl} means the dimension of the minimal coupled carrier supporting the required weak-type and color-type nonabelian charges on a common block. It is not the abstract minimal faithful representation dimension of the final gauge group.

Definition 7.2 (Weak-type and color-type nonabelian roles)**.** *A weak-type nonabelian charge is a pseudoreal nonabelian doublet role. A color-type nonabelian charge is an intrinsically complex nonabelian role, meaning that its nonabelian factor itself acts by an irreducible representation not equivalent to its conjugate. Abelian twisting of a pseudoreal nonabelian irreducible representation does not count as a color-type role.*

The derivation of the Standard Model quotient separates into five steps:

1. existence of a compact reconstructed group;
2. identification of the minimal nonabelian sector content;
3. identification of the minimal coupled carrier;
4. uniqueness of the connected abelian factor once admitted on that carrier;

5. determination of product gauge structure up to finite quotient, then of the global finite quotient from the realized matter spectrum.

Lemma 7.3 (Minimal nonabelian sector content)**.** *Any admissible low-energy sector must contain both a weak-type pseudoreal nonabelian charge role and an intrinsically complex color-type nonabelian charge role.*

Proof. Light chiral matter requires left-handed multiplets and right-handed singlets to carry inequivalent gauge data. A pseudoreal doublet structure is the minimal way to realize the weak sector without immediately producing vectorlike masses. A genuinely complex nonabelian representation is required to distinguish quarks from antiquarks. A single nonabelian simple factor cannot simultaneously supply both a minimal weak-type role and an intrinsically complex color-type role; abelian twisting does not count toward the color-type requirement. □

Lemma 7.4 (Connected 2D pseudoreal image classification)**.** *Let H be a connected compact group with a faithful irreducible* 2*-dimensional pseudoreal unitary representation. Then the connected derived subgroup of its image is conjugate to* $\mathrm{SU}(2)$*. Equivalently, the nonabelian factor acting in that role is* $\mathrm{SU}(2)$ *up to finite central quotient.*

Proof sketch. Irreducibility places the image inside $\mathrm{U}(2)$ with scalar commutant. Its connected derived subgroup is therefore a connected compact subgroup of $\mathrm{SU}(2)$. The connected compact subgroups of $\mathrm{SU}(2)$ are either tori or $\mathrm{SU}(2)$ itself. The representation is nonabelian and pseudoreal, so the torus case is excluded. Hence the derived subgroup is conjugate to $\mathrm{SU}(2)$, with only a finite central kernel left invisible in the representation. □

Lemma 7.5 (Connected 3D intrinsically complex image classification)**.** *Let H be a connected compact group with a faithful irreducible intrinsically complex* 3*-dimensional unitary representation. Then the connected derived subgroup of its image is conjugate to* $\mathrm{SU}(3)$*. Equivalently, the nonabelian factor acting in that role is* $\mathrm{SU}(3)$ *up to finite central quotient.*

Proof sketch. Because the representation is irreducible on $\mathbb{C}^3$, the semisimple part of the Lie algebra of the image acts irreducibly on $\mathbb{C}^3$. A product of two nontrivial simple factors would force a tensor-product decomposition of dimension at least 4, so only one simple factor can occur. Among compact simple Lie algebras, the only ones with nontrivial irreducible representations of dimension at most 3 are $\mathfrak{su}(2)$ and $\mathfrak{su}(3)$. The 3-dimensional $\mathfrak{su}(2)$ representation is real, not intrinsically complex, whereas the fundamental $\mathfrak{su}(3)$ representation is intrinsically complex. Therefore the connected derived subgroup is conjugate to $\mathrm{SU}(3)$, again up to finite central kernel. □

Lemma 7.6 (Minimal coupled carrier under MAR^+)**.** *Within the connected positive-dimensional Lie admissible class used by MAR, and under the weak-type / intrinsically complex color-type criteria above, the minimal coupled carrier supporting both roles on a common block has the form*

$$V = \mathbb{C}^3 \otimes \mathbb{C}^2, \qquad \chi_{\mathrm{cpl}} = 6.$$

Proof. By Lemma 7.4, the minimal weak-type nonabelian role is the pseudoreal doublet of $\mathrm{SU}(2)$, hence has dimension 2. By Lemma 7.5, the minimal intrinsically complex color-type nonabelian role is the triplet of $\mathrm{SU}(3)$, hence has dimension 3.

By definition of χ_{cpl}, the weak-type and color-type roles must act nontrivially on one common irreducible nonabelian block rather than on disconnected direct-sum summands. Commuting irreducible actions on such a common block therefore require dimension at least the product of the minimal weak and color dimensions:

$$\chi_{\mathrm{cpl}} \geq 2 \cdot 3 = 6.$$

Equality is achieved by the tensor-product carrier $\mathbb{C}^3 \otimes \mathbb{C}^2$. The block-diagonal representation $\mathbb{C}^3 \oplus \mathbb{C}^2$ of $S(U(3) \times U(2))$ is faithful of dimension 5, but it is not coupled and hence is not the MAR minimizer. Connected non-semisimple counterexamples of $U(2)$ type are excluded by the explicit color-type criterion rather than being silently ignored. □

Remark 7.7. *The dimension comparisons used in Lemma 7.6 apply to the positive-dimensional connected Lie image carrying the admissible nonabelian charges, equivalently to the identity component of the reconstructed group on that sector. Finite, disconnected, or connected non-semisimple counterexamples exist, but they are outside the admissible class fixed by the explicit weak-type / color-type criteria.*

MAR supplies existence of a connected abelian factor; the next lemma proves only uniqueness and identification on the minimal coupled carrier.

Lemma 7.8 (Uniqueness of the connected abelian factor on the minimal coupled carrier)**.** *Inside* $\mathrm{U}(6)$*, the commutant of* $\mathrm{SU}(3) \times \mathrm{SU}(2)$ *acting on* $\mathbb{C}^3 \otimes \mathbb{C}^2$ *is exactly* $\mathrm{U}(1)$*.*

Proof. The SU(3) action is irreducible on the first tensor factor and trivial on the second. The SU(2) action is irreducible on the second tensor factor and trivial on the first. By Schur's lemma, any operator commuting with both actions is scalar on each irreducible factor. Hence the full commutant is a single U(1). □

Theorem 7.9 (Product Gauge Structure up to Finite Quotient)**.** *Assume Axiom 3.5, vanishing obstruction* $[z] = 0$ *on the realized low-energy branch, the hypotheses of Theorem 7.1, the weak-type / color-type MAR premises above, and that the realized connected gauge image acts faithfully on the minimal coupled carrier. Then the realized connected gauge structure has the form*

$$G_{\mathrm{phys}} = \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\Gamma}$$

for some finite central subgroup Γ*.*

Proof. Theorem 7.1 yields some compact group G. Lemma 7.3 identifies the minimal admissible sector content, Lemmas 7.4 and 7.5 identify the connected nonabelian factors realizing the minimal weak-type and color-type roles, and Lemma 7.6 fixes the minimal coupled carrier. The tensor-product structure of that carrier implies commuting weak and color actions, so the connected semisimple part of the realized image is locally $\mathrm{SU}(3) \times \mathrm{SU}(2)$. Axiom 3.5 supplies the existence of one connected abelian charge factor acting nontrivially on the coupled carrier, and Lemma 7.8 shows that any such connected abelian factor is necessarily a single U(1). A compact connected Lie group with Lie algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ is a quotient of $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ by a finite central subgroup. Faithfulness rules out an invisible extra torus, so the only remaining ambiguity is the finite central quotient Γ. □

Theorem 7.10 (Hypercharge lattice on the realized matter package)**.** *Assume gauge group* $\mathrm{SU}(N_c) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y$*, one generation of chiral matter* (Q, u^c, d^c, L, e^c)*, one Higgs doublet* H*, and Yukawa terms*

$$QHu^c, \qquad QH^\dagger d^c, \qquad LH^\dagger e^c.$$

Then anomaly cancellation and Yukawa invariance imply

$$Y_L = -N_c Y_Q, \quad Y_H = N_c Y_Q, \quad Y_u = -(N_c + 1) Y_Q, \quad Y_d = (N_c - 1) Y_Q, \quad Y_e = 2 N_c Y_Q.$$

Fixing the normalization by $Q = T_3 + Y$ and $Q(\nu_L) = 0$ gives

$$Y_Q = \frac{1}{2N_c}.$$

For $N_c = 3$ one recovers the exact Standard Model lattice

$$Y_Q = \frac{1}{6}, \quad Y_L = -\frac{1}{2}, \quad Y_u = -\frac{2}{3}, \quad Y_d = \frac{1}{3}, \quad Y_e = 1, \quad Y_H = \frac{1}{2}.$$

Proof. Yukawa invariance gives

$$Y_u = -(Y_Q + Y_H), \qquad Y_d = -Y_Q + Y_H, \qquad Y_e = -Y_L + Y_H.$$

The mixed anomalies yield

$$N_c Y_Q + Y_L = 0,$$

and

$$2N_c Y_Q + N_c Y_u + N_c Y_d + 2Y_L + Y_e = 0.$$

Solving the first anomaly gives

$$Y_L = -N_c Y_Q.$$

Substituting the Yukawa relations and $Y_L = -N_c Y_Q$ into the mixed gravitational anomaly then yields

$$Y_H = N_c Y_Q, \qquad Y_u = -(N_c + 1)Y_Q, \qquad Y_d = (N_c - 1)Y_Q, \qquad Y_e = 2N_c Y_Q.$$

With these relations in place, the $\mathrm{SU}(N_c)^2\mathrm{U}(1)$ and $\mathrm{U}(1)^3$ anomalies cancel identically. The normalization of Y_Q is fixed by the electric charge operator $Q = T_3 + Y$ together with $Q(\nu_L) = 0$, which implies

$$Y_L = -\frac{1}{2} = -N_c Y_Q, \qquad \text{hence} \qquad Y_Q = \frac{1}{2N_c}.$$

□

Corollary 7.11 (Three Generations on the realized MAR-admissible branch)**.** *Under the hypotheses of Theorem 7.9,*

$$N_g = 3.$$

Proof. Intrinsic CKM CP violation requires

$$\frac{(N_g - 1)(N_g - 2)}{2} > 0,$$

hence $N_g \geq 3$. For one Higgs doublet, the one-loop weak-sector coefficient is

$$b_2 = \frac{11}{3} C_A - \frac{2}{3} \sum_f T(R_f) - \frac{1}{3} \sum_s T(R_s) = \frac{22}{3} - \frac{N_g(N_c + 1)}{3} - \frac{1}{6} = \frac{43}{6} - \frac{N_g(N_c + 1)}{3}.$$

Asymptotic freedom therefore implies

$$N_g(N_c + 1) < \frac{43}{2}.$$

Since the color-type requirement forces $N_c \geq 3$, this gives $N_g \leq 5$. The minimal admissibility selector then chooses $N_g = 3$. □

Corollary 7.12 (Three Colors on the realized MAR-admissible branch)**.** *Under the hypotheses of Theorem 7.9,*

$$N_c = 3.$$

Proof. Witten's global SU(2) anomaly requires the total number of left-handed weak doublets to be even:

$$N_g(N_c + 1) \equiv 0 \pmod 2.$$

With $N_g = 3$, this implies N_c is odd. The case $N_c = 1$ is inadmissible because it removes the intrinsically complex color sector demanded by Lemma 7.3. Minimal admissibility therefore selects $N_c = 3$ [12]. □

Proposition 7.13 (Global quotient from the realized matter spectrum)**.** *If the realized matter spectrum carries the hypercharges of Theorem 7.10 with $N_c = 3$, then the subgroup of $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ acting trivially on all realized states is exactly $\mathbb{Z}_6$.*

Proof. Write $q := 6Y \in \mathbb{Z}$ and $\eta := e^{i\pi/3}$, so the U(1) factor acts by η^q on charge-q multiplets. The element

$$g_0 := (\omega_3, -1, \eta)$$

acts trivially on the realized matter and Higgs multiplets:

$$Q:\ \omega_3(-1)\eta = 1, \qquad u^c:\ \omega_3^{-1}\eta^{-4} = 1, \qquad d^c:\ \omega_3^{-1}\eta^2 = 1,$$
$$L:\ (-1)\eta^{-3} = 1, \qquad e^c:\ \eta^6 = 1, \qquad H:\ (-1)\eta^3 = 1.$$

Its powers therefore generate a subgroup of order six acting trivially on all realized states. Conversely, any central element acting trivially on e^c must have U(1) part η^n, and triviality on Q then fixes the SU(3) and SU(2) center factors uniquely. Hence every trivial central element is a power of g_0, so the kernel is exactly $\mathbb{Z}_6$. □

Corollary 7.14 (Standard Model Gauge Group)**.** *Under Theorem 7.9, Theorem 7.10, Corollaries 7.11 and 7.12, and Proposition 7.13,*

$$G_{\text{phys}} = \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}.$$

Proof. Theorem 7.9 yields the product structure up to finite quotient, and Proposition 7.13 fixes that quotient to $\mathbb{Z}_6$ once the realized hypercharge lattice and $N_c = 3$ are in place. □

Corollary 7.15 (Product-Group Consequences)**.** *The adjoint representation of the full connected gauge group contains only*

$$(8, 1, 0) \oplus (1, 3, 0) \oplus (1, 1, 0).$$

Equivalently, the adjoint representation of the derived gauge group contains only

$$(8, 1, 0) \oplus (1, 3, 0).$$

There are no mixed $(3, 2, \pm 5/6)$ gauge bosons. Hence gauge-mediated proton decay is absent. The unbroken gauge symmetries forbid elementary gauge-boson mass terms: the photon remains massless, while gluons are massless gauge fields but confined.

Remark 7.16. *The same product-group structure removes the usual grand-unification monopole sector. In the present framework, unification of couplings does not proceed by embedding the gauge group into a simple group that is later broken. It proceeds through a shared geometric origin of the edge heat-kernel parameters. Absence of gauge-mediated proton decay and unification therefore coexist without tension.*

8 Two-Input Quantitative Sector

8.1 Continuous-input count in the current quantitative implementation

The quantitative program uses the same axiom set together with the two external continuous inputs P and N_{scr} in its current supplement-backed numerical implementation. In the current implementation, P is calibrated from the gauge sector and N_{scr} is inferred from the observed cosmological constant. The discrete structure

$$(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6, \quad N_c = 3, \quad N_g = 3, \quad \varepsilon = \frac{1}{6}, \quad \delta = \frac{2}{9}$$

is then fixed by structural arguments, MAR selection, or phenomenological continuation steps rather than by direct fits. On the structural, calibration, and Higgs/top branch alone, this replaces roughly nineteen particle-physics inputs, or twenty-six when neutrino parameters are included, by two external continuous inputs plus structural relations. Later flavor continuations use additional phenomenological branch choices and are not part of that count.

8.2 Epistemic classification of outputs

The quantitative sector is not uniform. The paper uses four categories.

1. **Calibration-sector consistency checks.** These are algebraically entangled with the determination of P from the gauge sector. They include v, the gauge couplings at m_Z, and tree-level electroweak masses.

2. **Independent quantitative outputs.** These use additional structural conditions and add no further continuous fit once the gauge trajectory and scale-setting branch are fixed. In the compact note this category is the Higgs/top critical-surface branch.

3. **Parameter-free structural outputs.** These include the gauge quotient, the exact hypercharge lattice on the realized matter package, and $N_c = 3$, $N_g = 3$ on the realized MAR-admissible branch.

4. **Phenomenological continuations.** These include the charged-lepton/Koide construction, quark textures, baryogenesis, dark-sector phenomenology, and any use of the $\mathbb{Z}_6$ suppression factor as a flavor or entropy ansatz.

The quark-hierarchy and neutrino branches are not at the same maturity level in the compact note: the former is a texture-pattern continuation, while the latter is currently only an order-of-magnitude capacity-branch estimate.

This classification matters because the framework is not presented as a fit to all observables. Its stronger claim is that a large set of discrete and dimensionless structures are fixed before the overall energy scale is chosen.

8.3 Fundamental scales

The unification scale is

$$M_U = \frac{E_P}{e^{2\pi}} P^{1/6} \approx 2.474 \times 10^{16} \text{ GeV},$$

and the cell energy is

$$E_{\text{cell}} = \frac{E_P}{\sqrt{P}} \approx 9.56 \times 10^{18} \text{ GeV}.$$

The factor $e^{2\pi}$ comes from modular geometry. The factor $P^{1/6}$ comes from the cell-area scaling relation.

8.4 The pixel constraint and gauge closure

Edge-center completion and MaxEnt produce open-edge Gibbs / Laplacian weights

$$p_R(t) \propto d_R e^{-tC_2(R)}, \qquad \bar{\ell}_G(t) = \sum_R p_R(t) \log d_R.$$

For the two nonabelian factors, these are explicitly

$$\bar{\ell}_{\mathrm{SU}(2)}(t) = \sum_{j=0,\frac{1}{2},1,\ldots} p_j(t) \log(2j+1), \qquad p_j(t) \propto (2j+1) e^{-t\, j(j+1)},$$

and

$$\bar{\ell}_{\mathrm{SU}(3)}(t) = \sum_{p,q\geq 0} p_{(p,q)}(t) \log d_{(p,q)}, \qquad p_{(p,q)}(t) \propto d_{(p,q)} e^{-tC_2(p,q)},$$

with

$$d_{(p,q)} = \frac{1}{2}(p+1)(q+1)(p+q+2),$$

and

$$C_2(p,q) = \frac{1}{3}(p^2+q^2+pq+3p+3q).$$

Upon gluing or tracing the two boundary indices one obtains the closed heat-kernel partition function

$$Z_{\mathrm{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)}.$$

For the nonabelian factors one obtains the pixel constraint

$$\frac{P}{4} = \bar{\ell}_{\mathrm{SU}(2)}(t_2) + \bar{\ell}_{\mathrm{SU}(3)}(t_3), \qquad t_i = 4\pi^2 \alpha_i.$$

The framework then solves a supplement-backed self-consistent fixed-point problem:

1. choose a trial unified coupling α_U;
2. run the three gauge couplings to a trial scale using the derived beta-function shifts;
3. compute the electroweak scale

$$v = E_{\mathrm{cell}} \exp\left(-\frac{2\pi}{\beta_{\mathrm{EW}} \alpha_U}\right), \qquad \beta_{\mathrm{EW}} = N_c + 1 = 4;$$

4. impose $m_Z(\mu) = \mu$ at the running scale;
5. solve the pixel constraint by bisection in α_U.

This yields

$$\alpha_U \approx 0.04112, \qquad \alpha_U^{-1} \approx 24.32.$$

The resulting beta-function shifts are numerically close to the MSSM one-loop shifts. In the framework this is not introduced by hand. It comes from the same heat-kernel edge weighting that enters the pixel constraint. The compact note quotes this closure only at the level needed to state the numerical branch; the full beta-function shifts, threshold choices, and matching conventions remain supplement-backed.

8.5 Calibration-sector electroweak outputs

We use the convention

$$\alpha_1 = \frac{5}{3}\alpha_Y, \qquad g_Y^2 = 4\pi\alpha_Y = 4\pi\frac{3}{5}\alpha_1.$$

The gauge couplings at $\mu = m_Z$ are then

$$\alpha_1(m_Z) \approx 0.01696, \qquad \alpha_2(m_Z) \approx 0.03384, \qquad \alpha_3(m_Z) \approx 0.1183,$$

which, under the printed conventions of this compact note, give

$$\alpha_{\rm em}^{-1}(m_Z) \approx 127.82, \qquad \sin^2\theta_W(m_Z) \approx 0.23119.$$

With

$$v \approx 246.77 \text{ GeV},$$

the tree-level electroweak masses are

$$m_W = \frac{1}{2}vg_2 \approx 80.46 \text{ GeV},$$

and

$$m_{Z,\rm run} = \frac{1}{2}v\sqrt{g_2^2 + g_Y^2} \approx 91.76 \text{ GeV}.$$

These are calibration-sector outputs. Their role in the paper is scale setting, not independent confirmation. A fully reproducible numerical audit requires the beta-function shifts and matching conventions supplied in the repository supplements. The compact note therefore reports only the running-Z relation internally and does not treat an illustrative pole-mass estimate as a benchmark output.

Quantity	OPH output	Experimental value	Status
$\alpha_{\rm em}^{-1}(m_Z)$	127.82	127.952	calibration sector
$\sin^2\theta_W(m_Z)$	0.23119	0.23122	calibration sector
$\alpha_s(m_Z)$	0.1183	0.1179	calibration sector
m_W	80.46 GeV	80.38 GeV	calibration sector

8.6 Higgs and top from the critical surface

The independent quantitative branch begins with the refinement-stability condition

$$\lambda(M_U) = 0, \qquad \beta_\lambda(M_U) = 0.$$

At one loop, these conditions determine the top Yukawa boundary value

$$y_t(M_U) = \left[\frac{1}{16}\left(2g_2^4 + (g_2^2 + g_Y^2)^2\right)\right]^{1/4}.$$

The logic is simple: once $\lambda(M_U) = 0$, the condition $\beta_\lambda(M_U) = 0$ removes the final freedom in the top Yukawa boundary value. The low-energy Higgs and top masses then follow by RG evolution rather than by direct fitting.

Running the coupled system $(y_t, \lambda, g_Y, g_2, g_3)$ down to $\mu \sim m_t$ yields

$$m_t^{\overline{\rm MS}}(m_t) \approx 160.6 \text{ GeV},$$

and after the standard three-loop QCD pole-mass conversion used in the present implementation,

$$m_t^{\text{pole}} \approx 171.1 \text{ GeV}.$$

The corresponding Higgs mass is

$$m_H = \sqrt{2\lambda(m_t)}\, v \approx 126.5 \text{ GeV}.$$

The critical-surface boundary conditions are not an additional continuous fit once the gauge trajectory and scale-setting branch are fixed. Converting the resulting dimensionless ratios m_H/v and m_t/v to GeV still uses the previously calibrated v and gauge-coupling trajectory, and the critical-surface branch can fail even when the electroweak calibration succeeds.

Quantity	OPH output	Experimental value	Status
m_H	126.5 GeV	125.2 GeV	independent quantitative
m_t^{pole}	171.1 GeV	172.6 GeV	independent quantitative

9 Phenomenological Flavor Continuations

Theorem 3.10 packages the compact note's recovered core: the D1–D5 relativity chain together with the D7–D9 Standard Model structural chain. D6 is a separate input-dependent capacity corollary, and D10–D11 are supplement-backed numerical branches of the current implementation. Nothing in the present section is part of that recovered core. The role of the section is only to record which flavor ideas have been explored and exactly why they remain outside the proved package.

9.1 The $\mathbb{Z}_6$ suppression parameter

If one adds the extra assumption of a *uniform* sixfold center-label ensemble on the realized $\mathbb{Z}_6$ quotient sectors, then the associated center-label entropy is

$$\Delta S_{\text{center}} = \ln 6.$$

and one may define the phenomenological hierarchy parameter

$$\varepsilon = e^{-\Delta S_{\text{center}}} = \frac{1}{6}.$$

The crucial scope point is that the quotient

$$(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$$

does *not* by itself force the uniform ensemble. The step from the proved quotient to the numerical suppression factor $\varepsilon = 1/6$ is therefore an additional flavor ansatz, not a theorem of the recovered core.

9.2 Koide, charged-lepton, and texture branches

Every continuation in which one introduces a circulant generation-space ansatz, a fixed Koide phase such as

$$Q = \frac{2}{3}, \qquad \delta = \frac{2}{9},$$

or additional texture exponents depends on extra phenomenological structure beyond the recovered core. These constructions may be interesting as model-building continuations because they reuse

discrete data already fixed on the realized Standard Model branch, but they are not derived by overlap consistency, the scaling-limit gravity premises, compact gauge reconstruction, anomaly cancellation, or MAR alone.

Accordingly, the compact note does *not* claim the charged-lepton masses, the Koide relation, or texture exponents as recovered outputs. At present they should be read only as conjectural downstream organizations of flavor data. A failure of any such ansatz would retract that continuation, not Theorem 3.10.

9.3 Neutrino scale

The neutrino discussion is weaker still in scope. The present note contains only an input-dependent, order-of-magnitude capacity estimate tied to N_{scr}; it does not provide a benchmark neutrino mass formula, a mixing derivation, or a completed flavor mechanism. The estimate itself sits at the separate D6 boundary, while any fuller neutrino branch remains outside both the recovered core and the supplement-backed D10–D11 quantitative implementation.

Phase I therefore ends *before* flavor is derived. The compact paper recovers spacetime dynamics and Standard Model gauge/matter structure first; flavor remains an explicitly deferred problem rather than a claim hidden inside the packaging.

10 Program-Level Phenomenology and Further Continuations

Theorem 3.10 marks the end of the compact note's recovered core. The separate input-dependent capacity corollary D6 and the D10–D11 numerical branches are kept visible, but the items below go beyond that scope. They are recorded only to show where the broader OPH program might go next; they are *not* theorem-level outputs of the compact paper.

10.1 What is already fixed before any continuation is added

Two points should be separated from the downstream branches. First, the product-group consequence

$$\tau_p^{(\mathrm{gauge})} = \infty$$

for gauge-mediated proton decay is already a corollary of the realized Standard Model structure and does not depend on any dark-sector, baryogenesis, string, or spectroscopy continuation. Second, the relation

$$\Lambda = \frac{3\pi}{G N_{\mathrm{scr}}}$$

is already isolated earlier as the separate input-dependent capacity corollary D6 once the screen-capacity identification is adopted. Neither point should be confused with the conjectural branches below.

10.2 Dark-sector branch

The modular-anomaly proposal for an effective dark stress tensor requires more than the recovered gravity branch. The compact note does derive the modular additivity defect

$$\Delta K_\delta = K_{ABD} - K_{AB} - K_{BD} + K_B$$

as a recoverability diagnostic, but it does not prove that a nontrivial cap-local anomaly

$$K_C^{(\mathrm{anom})} := K_C - 2\pi B_C$$

survives with the sign, magnitude, and infrared closure required to explain galaxy-scale phenomenology. Any MOND/RAR-type acceleration law extracted from this idea therefore remains a conjectural response model, not a compact-note prediction.

10.3 Baryogenesis, spectroscopy, and string/worldsheet branches

The same scope rule applies to the other downstream topics. Baryogenesis estimates built from powers of a phenomenological suppression factor, black-hole spectroscopy templates built from an additional discrete-horizon branch, and large-N_{edge} string/worldsheet reorganizations all require premises that are not established by the end of Theorem 3.10. In particular:

1. baryogenesis needs a concrete out-of-equilibrium mechanism and a justified defect-counting dynamics;
2. black-hole spectroscopy needs a discrete-horizon branch with its own spacing law and selection rules;
3. string/worldsheet claims need a controlled large-N_{edge} regime distinct from the physical color rank $N_c = 3$.

Until those extra premises are proved, these subjects remain program-level continuations only.

10.4 Reading rule for the compact note

The compact note should therefore be read with the following boundary in mind: relativity and Standard Model structure are the recovered core; the D6 capacity relation and the supplement-backed D10–D11 quantitative implementation are secondary; dark-sector, baryogenesis, spectroscopy, and string/worldsheet discussions are deferred continuations whose success or failure does not by itself confirm or refute the recovered core.

11 Relation to Holography and Existing UV Frameworks

OPH belongs to the holographic family of ideas, but its primitive data are different from those used in asymptotic-boundary constructions. The distinction matters because several of the central claims of the present paper, including the treatment of Λ, the handling of factorization, and the emergence of gauge structure, depend on it.

11.1 Static-patch screen versus asymptotic boundary

In asymptotic-boundary holography one starts from a dual theory at infinity and reconstructs the bulk inward. In OPH one starts from a finite-capacity screen equipped with a net of local patch algebras and reconstructs the effective bulk from overlap consistency. The primitive object is therefore a family of observer patches with nontrivial overlaps, not a single global boundary theory.

This shift has two technical consequences. First, subsystem factorization is treated from the beginning as a gluing problem with centers and sector labels. Second, the physical role of the horizon is local and operational: every observer has direct access only to a patch algebra, and global law is whatever survives reconciliation on overlaps.

11.2 Positive Λ and finite capacity

The null-modular reconstruction of Section 6 determines the stress tensor only up to a metric term. This means that local null data do not fix Λ. The cosmological constant enters only when one supplies the global screen capacity

$$N_{\mathrm{scr}} = \log \dim \mathcal{H}_{\mathrm{tot}}.$$

The resulting relation

$$\Lambda = \frac{3\pi}{G N_{\mathrm{scr}}}$$

fits naturally with a de Sitter static patch of finite entropy via $N_{\mathrm{scr}} = A_{\mathrm{dS}}/(4G)$. OPH is therefore formulated in a de Sitter-first language. The positive cosmological constant is tied to finite capacity rather than to a deformation of a negative-Λ starting point.

11.3 Factorization, gauge structure, and the string sector

The factorization problem of gauge theory and gravity is often treated as a technical nuisance. In OPH it is part of the architecture. Edge-center completion turns the collar center into a first-class object, and the entropy split

$$S_{\mathrm{gen}} = \langle L_C \rangle + S_{\mathrm{bulk}}$$

is then a structural consequence rather than an added prescription.

In the bosonic EFT branch, the same collar data produce a symmetric rigid transportable sector category. Compact gauge reconstruction then follows conditionally from that category, and the Standard Model quotient is selected by minimal admissibility. This differs sharply from approaches in which the gauge group is specified in advance.

The string-theory relation is similarly reversed. The edge partition function

$$Z_{\mathrm{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)}$$

is the closed partition function obtained after gluing the open-edge weights $p_R(t) \propto d_R e^{-tC_2(R)}$. It has the heat-kernel form familiar from two-dimensional Yang-Mills theory. If a large-N_{edge} regime exists, with N_{edge} distinct from the physical color rank $N_c = 3$, its free energy admits a genus expansion. Within OPH this is read as a conditional effective reorganization of edge degrees of freedom. String theory therefore enters as a sector description, while the microscopic starting data are the screen net, overlap consistency, and sector reconstruction.

Feature	Asymptotic-boundary holography	OPH
Primitive data	Boundary theory at infinity	Finite-capacity screen with observer-patch net
Factorization across cuts	Subtle and often indirect	Explicit edge-center completion with central labels
Status of Λ	External background datum	Fixed globally by N_{scr} after local null reconstruction
Gauge group	Usually specified as part of the model	Reconstructed from edge sectors, then filtered by admissibility
String sector	Fundamental dual description	Effective large-N_{edge} reorganization of edge dynamics

12 Predictive Status and Falsifiability

This section is the authoritative falsifiability ledger for the three-phase program. In the compact note, the recovered core is

$$(D1\text{–}D5) \cup (D7\text{–}D9),$$

namely the relativity chain together with the realized Standard Model structural chain. D6 is a separate input-dependent capacity corollary, D10–D11 are implementation-level numerical branches of the current supplement-backed realization, and D12 collects deferred continuations.

12.1 Prediction audit by claim tier

Tier	Representative outputs	What would actually falsify the tier
Phase I recovered core (Theorem 3.10; D1–D5, D7–D9)	Confluence of overlap repair, conditional Lorentz kinematics, the conditional Jacobson-type Einstein branch in the stated scaling regime, compact gauge reconstruction in the bosonic branch, the realized Standard Model quotient chain, the exact hypercharge lattice on the realized matter package, $N_c = 3$, $N_g = 3$, and the product-group consequence of no gauge-mediated proton decay	A mathematical failure in the derivation chain, or data that require a different realized gauge quotient, different realized hypercharges, a different color or generation count on the admissible branch, or gauge-mediated proton decay
Phase II input-dependent corollary (D6)	$\Lambda = \frac{3\pi}{GN_{\mathrm{scr}}}$ once the screen-capacity identification is made, together with the separate order-of-magnitude neutrino estimate tied to the same capacity input	Failure of the screen-capacity identification or of the specific global capacity relation. Such a failure would not by itself erase the recovered core
Phase II current implementation: calibration sector (D10)	Pixel-closure gauge/electroweak consistency, $\alpha_i(m_Z)$, $\alpha_{\mathrm{em}}^{-1}(m_Z)$, $\sin^2\theta_W(m_Z)$, v, and tree-level electroweak masses	Failure of the supplement-backed closure of the present implementation once P and the printed running/matching conventions are imposed
Phase II current implementation: secondary quantitative branch (D11)	Higgs/top critical-surface branch	Failure of the extra critical-surface condition or of the supplement-backed RG transport used in that branch; such a failure does not by itself erase the recovered core
Phase III deferred continuations (D12 and beyond)	Flavor ansätze, charged-lepton/Koide fits, texture branches, neutrino mass/mixing constructions beyond the D6 estimate, dark-sector response laws, baryogenesis estimates, black-hole spectroscopy templates, string/worldsheet reorganizations, and other downstream phenomenology	Retraction of the corresponding continuation only. These branches are not part of the compact paper's recovered core and their failure is not, by itself, a falsification of relativity-plus-Standard-Model recovery

This tier split is the sync rule for mirrored status surfaces: Phase I is the recovered-core falsifiability standard, Phase II is visible but non-core, and Phase III is explicitly continuation-level.

12.2 Recovered-core discriminators

Once the claim boundary is made explicit, the sharpest discriminators are the ones that do not depend on auxiliary flavor, dark-sector, baryogenesis, or string ansätze.

12.2.1 Realized Standard Model gauge structure

On the realized MAR-admissible branch, the compact note fixes

$$G_{\text{phys}} = \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_c = 3, \qquad N_g = 3,$$

with the exact one-generation hypercharge lattice

$$Y_Q = \frac{1}{6}, \qquad Y_L = -\frac{1}{2}, \qquad Y_u = -\frac{2}{3}, \qquad Y_d = \frac{1}{3}, \qquad Y_e = 1, \qquad Y_H = \frac{1}{2}.$$

Evidence that the realized low-energy gauge structure or realized hypercharges differ from these values would directly contradict the recovered core.

12.2.2 Product-group corollary

Because the realized gauge structure is a product group up to the finite central quotient already fixed above, there are no mixed $(3, 2, \pm 5/6)$ gauge bosons. Hence

$$\tau_p^{(\text{gauge})} = \infty$$

for gauge-mediated proton decay. Observation of proton decay specifically attributable to gauge-boson exchange would therefore falsify the realized product-group branch.

12.2.3 Relativity branch

The relativity claim is conditional but still sharp about its scope: if the stated geometric-modular, null-bridge, and fixed-cap-stationarity premises are realized in the intended scaling regime, then the compact note predicts Lorentz kinematics on the screen and a Jacobson-type Einstein branch. What is being tested here is the existence of that scaling regime and its premises, not a fixed-cutoff matrix-algebra identity.

12.3 What does *not* count as a contradiction of the recovered core

The following do *not* falsify Theorem 3.10 if they fail in their present form:

1. the uniform $\mathbb{Z}_6$ center-label ensemble and the associated $\varepsilon = 1/6$ flavor ansatz;
2. Koide/charged-lepton constructions, texture exponents, or capacity-level neutrino estimates;
3. modular-anomaly dark-sector response laws or MOND/RAR-style scaling claims;
4. baryogenesis estimates based on extra suppression-counting assumptions;
5. discrete-horizon spectroscopy templates;
6. large-N_{edge} string/worldsheet reorganizations.

These are all post-Phase-I, non-core branches. Some are Phase-II input-dependent estimates, while the rest are Phase-III continuations. None is part of the compact note's recovered relativity-plus-Standard-Model core.

12.4 What the compact paper claims after scope-cleaning

After the present scope clean, the compact paper claims exactly this: observer-overlap consistency, together with the stated scaling-limit and categorical premises, recovers a conditional relativity branch and the realized Standard Model structural branch. The Phase-II capacity relation and the current Phase-II D10–D11 implementation remain visible but secondary. Everything in Phase III is explicitly non-core.

13 Common Objections and Clarifications

13.1 Is the use of P circular?

The answer depends on which sector is under discussion. In the current implementation the pixel area P calibrates the overall particle-physics scale through the gauge sector. Quantities that are algebraically entangled with that calibration, such as v, m_W, m_Z, and the gauge couplings at m_Z, are consistency checks. The paper treats them as such.

The stronger outputs enter only after the scale has been set. The critical-surface condition

$$\lambda(M_U) = 0, \qquad \beta_\lambda(M_U) = 0$$

fixes the Higgs and top branch without introducing a new continuous fit beyond the calibrated gauge trajectory and scale-setting branch. The Koide continuation is controlled by

$$Q = \frac{2}{3}, \qquad \delta = \frac{2}{9},$$

which are continuation inputs tied to the previously derived discrete data inside that branch. The quark hierarchy depends on the $\mathbb{Z}_6$ texture ansatz, and the neutrino branch depends on the second input N_{scr}, not on P.

Operationally, changing P moves the calibration family $(\alpha_U, \alpha_i(m_Z), v, m_W, m_Z)$ as well as the overall GeV scale. It does not alter the parameter-free structural outputs such as the gauge quotient or hypercharge lattice. The useful distinction is therefore between scale setting, structural determination, and later phenomenological continuations.

13.2 Does a fixed UV cell size break Lorentz invariance?

The UV regulator is placed on the screen algebra, not on an emergent bulk spacetime lattice. Lorentz kinematics in the present framework arise from the continuum modular action on caps, summarized in Corollary 6.3, rather than from microscopic invariance of a bulk discretization.

This is conceptually similar to other continuum limits in statistical mechanics and lattice field theory. A regulator may break a symmetry at finite cutoff while the infrared fixed point restores it exactly. In OPH the restoration is stronger than a generic RG expectation because the modular-flow theorem identifies the symmetry group itself:

$$\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1).$$

Residual Lorentz violation would therefore have to arise from a failure of the regulator premises or from a failure of the modular-flow theorem, not merely from the existence of a UV cell size.

13.3 Why use type-I algebras if continuum QFT uses type III?

The type-I assumption is a regulator statement. At finite cutoff the local patch algebras are finite-dimensional matrix algebras, which makes entropy and recovery theory well defined. The continuum limit is expected to recover the familiar type-III behavior of local QFT algebras. There is no contradiction here. The two statements refer to different scales.

This is no more problematic than using a lattice regulator for Yang-Mills theory and then taking a continuum limit in which the operator algebra changes type. The mathematical burden is the existence of a controlled scaling regime. In the present manuscript that burden is isolated in Assumption 3.6. It is not hidden inside the main theorems.

13.4 Why is charge quantization possible without a simple GUT?

The framework uses the global quotient and anomaly structure rather than a simple-group embedding. Once the realized matter content is fixed, the subgroup acting trivially on all states is exactly $\mathbb{Z}_6$, and the anomaly equations force the sixth-integer hypercharge lattice. Integer electric charge for color singlets is then a property of the realized quotient structure. A simple unified gauge group is not required.

14 Discussion

The framework is strongest where the outputs are discrete or structurally rigid:

1. the Lorentz branch inside the assumed geometric modular phase;
2. the conditional scaling-limit Einstein branch under fixed-cap stationarity and the null-stress premise;
3. the Standard Model gauge quotient chain;
4. exact hypercharges on the realized one-generation matter package with one Higgs doublet;
5. three colors and three generations on the realized MAR-admissible branch;
6. the compact two-input economy of the present quantitative implementation, kept explicit as a secondary layer rather than as part of Phase I.

The next strongest sector is the secondary quantitative D11 branch consisting of the Higgs and top masses. The separate D6 cosmological-capacity relation remains visible as an input-dependent corollary rather than as part of the recovered core. The charged-lepton and Koide sectors are promising but should presently be read as phenomenological continuations rather than core theorem outputs.

The weakest part is the light-quark and hadron branch. The hierarchy pattern is robust, but precision control of low-energy QCD effects, scheme dependence, and order-one coefficients is not yet complete.

The framework should therefore be evaluated in stages. It already contains several results that are normally postulated independently in other approaches: kinematics, dynamics, gauge structure, charge lattice, matter multiplicities, and part of the mass spectrum in supplement-backed quantitative form. Whether one regards this as evidence for fundamentality depends on whether these simultaneous derivations survive further scrutiny.

Topics intentionally excluded from the present formulation include simulation narratives, observer-continuation mechanisms, and claims of final completeness for all low-energy observables.[2]

15 Conclusion

Observer-Patch Holography starts from overlap consistency of observer patches and yields, within one framework, several effective structures usually treated separately in modern fundamental physics. The compact note supports a schedule-independent normal form for objective law, a conditional Lorentz branch in the assumed geometric modular phase, a conditional Jacobson-type Einstein branch under fixed-cap stationarity and the null-stress premise, the Standard Model gauge quotient with exact hypercharges on the realized matter package under the extended MAR admissibility package in the bosonic internal-gauge branch, three colors and three generations on that realized branch, a current two-input quantitative implementation whose supplement-backed Higgs/top branch is secondary to the recovered core, and a separate input-dependent cosmological-capacity corollary. It also records several phenomenological continuations whose present status is weaker and is labeled as such in the text.

Accordingly, the framework functions as a candidate underlying theory from which general relativity and the Standard Model may arise as effective descriptions once the stated scaling-limit and categorical premises are realized. The next technical tasks are sharper quantitative control, a fuller fermionic gauge branch, and explicit UV completions that land in the required modular/geometric phase.

A One-loop RG Equations Used in the Quantitative Branch

For convenience we record the one-loop system used in the Higgs/top transport branch: Throughout this appendix, g_Y denotes the SM hypercharge coupling. The GUT-normalized coupling is g_1 with $g_1^2 = \frac{5}{3} g_Y^2$, equivalently $\alpha_1 = \frac{5}{3}\alpha_Y$. Therefore the one-loop coefficient is 41/6 in the g_Y equation and 41/10 in the g_1 equation.

$$16\pi^2 \frac{dg_Y}{d\ln\mu} = \frac{41}{6} g_Y^3, \tag{3}$$

$$16\pi^2 \frac{dg_2}{d\ln\mu} = -\frac{19}{6} g_2^3, \tag{4}$$

$$16\pi^2 \frac{dg_3}{d\ln\mu} = -7 g_3^3, \tag{5}$$

$$16\pi^2 \frac{dy_t}{d\ln\mu} = y_t \left(\frac{9}{2} y_t^2 - \frac{17}{12} g_Y^2 - \frac{9}{4} g_2^2 - 8 g_3^2 \right), \tag{6}$$

$$16\pi^2 \frac{d\lambda}{d\ln\mu} = 24\lambda^2 + \lambda \left(12 y_t^2 - 9 g_2^2 - 3 g_Y^2 \right) - 6 y_t^4 + \frac{3}{8} \left(2 g_2^4 + (g_2^2 + g_Y^2)^2 \right). \tag{7}$$

In the present supplement-backed numerical implementation this system is the transport layer between the UV boundary data and the low-energy Higgs/top outputs. Numerically, the compact note uses one-loop $\overline{\mathrm{MS}}$ transport, the matching scale $\mu = m_Z = 91.1876\,\mathrm{GeV}$, the unified scale $M_U \approx 2.474 \times 10^{16}\,\mathrm{GeV}$ from the main text, and the standard three-loop QCD pole-mass conversion for the displayed top-branch number. Higher-loop and threshold refinements are expected to shift the final numbers at the GeV level rather than alter the logic of the branch.

[2] Questions outside this theorem package are, for present purposes, mostly harmless.

B Quantitative Robustness Audit

Source of variation	Baseline used here	Expected effect on outputs
Critical-surface RG transport	one-loop SM running	moving to two loops is expected to shift m_H by about 1 GeV and m_t by about 0.5 GeV
Pole-mass and threshold matching	standard three-loop top conversion plus leading electroweak corrections	order-one GeV effects in the top branch; sub-GeV to GeV effects in the Higgs branch
Heat-kernel truncation in $\bar{\ell}_G$	200 representations	negligible at displayed precision, with changes below 10^{-12} in the entropy sums
Bisection tolerance for the pixel closure	10^{-10}	affects only the last shown digits of α_U and derived calibration outputs
Light-quark scheme matching	not yet implemented at precision level	dominant source of the present 5% to 20% uncertainty in the light-quark branch

The paper therefore separates the quantitative sector by status. The Higgs and top branches are useful because their errors are controlled at the percent level. The charged-lepton continuation is numerically sharp but rests on a weaker discrete-selection step, while the light-quark and hadronic branch still carries the largest scheme and threshold uncertainty.

References

[1] J. J. Bisognano and E. H. Wichmann, "On the duality condition for a Hermitian scalar field," *J. Math. Phys.* **16** (1975) 985–1007.

[2] J. J. Bisognano and E. H. Wichmann, "On the duality condition for quantum fields," *J. Math. Phys.* **17** (1976) 303–321.

[3] E. H. Lieb and M. B. Ruskai, "Proof of the strong subadditivity of quantum-mechanical entropy," *J. Math. Phys.* **14** (1973) 1938–1941.

[4] D. Petz, "Sufficient subalgebras and the relative entropy of states of a von Neumann algebra," *Commun. Math. Phys.* **105** (1986) 123–131.

[5] D. Petz, "Sufficiency of channels over von Neumann algebras," *Quart. J. Math.* **39** (1988) 97–108.

[6] O. Fawzi and R. Renner, "Quantum conditional mutual information and approximate Markov chains," *Commun. Math. Phys.* **340** (2015) 575–611.

[7] P. Hayden, R. Jozsa, D. Petz and A. Winter, "Structure of states which satisfy strong subadditivity of quantum entropy with equality," *Commun. Math. Phys.* **246** (2004) 359–374.

[8] T. Jacobson, "Thermodynamics of spacetime: the Einstein equation of state," *Phys. Rev. Lett.* **75** (1995) 1260–1263.

[9] T. Jacobson, "Entanglement equilibrium and the Einstein equation," *Phys. Rev. Lett.* **116** (2016) 201101.

[10] S. Doplicher and J. E. Roberts, "A new duality theory for compact groups," *Invent. Math.* **98** (1989) 157–218.

[11] S. Doplicher and J. E. Roberts, "Why there is a field algebra with a compact gauge group describing the superselection structure in particle physics," *Commun. Math. Phys.* **131** (1990) 51–107.

[12] E. Witten, "An SU(2) anomaly," *Phys. Lett. B* **117** (1982) 324–328.